

Research Methodology

Course Code: 21IPC501J²

COURSE CONTAINS

Module-1 – RESEARCH PREPARATION, PLANNING AND RESOURCES

16 Hours

Research – objectives, definitions – knowledge economy; Perspectives of stakeholders; Types of research; Critical & creative thinking; Generating research topic; Identifying broad research question and objectives – design of questionnaire; Characteristics of a good research – research design; Methods of scientific enquiry – theoretical, experimental and empirical; Hypothesis – formulation & testing. Sources of information – ICT enabled tools; Databases, repositories, public and private sources, indexes; Literature search keyword; Backward and forward search; Quality measurement tools: journal metrics, author-level metrics; Meta-analysis of research findings; Literature review: grouping, analysing & comparison: Identification of research Methods; Intellectual Property Rights; Copyright – patent –geographical indication – industrial designs – technology transfer. PRACTICE: Identifying research topic and formulation of Research Statement; Conduction of literature review.

Module-2 – ACADEMIC WRITING AND PRESENTATIONS

14 Hours

Preparation of project proposal for funding – Identification of funding agencies; Project proposals structure; Research report writing; Data visualization; Communication model audience analysis; Identification of suitable journal; Conference presentation types: Oral, poster – difference in audience interaction – guideline for effective presentation; Synopsis thesis – extended abstract – graphical and video abstract – short communications; Referencing – style – tools for referencing; Thesis writing: structure – preliminary pages, main body, references; Evaluation of Thesis – examiner reports – Oral defence. PRACTICE: Writing abstract and conclusion; Referencing in different styles.

Module-3 – PUBLICATION ETHICS AND MISCONDUCTS

12 Hours

Ethics: definition; Ethics with respect to science and research - intellectual honesty and research integrity; Scientific misconducts: Falsification, Fabrication and Plagiarism (FFP); Redundant publications: duplicate and overlapping publications, salami slicing; Selective reporting and misrepresentation of data; Publication ethics: introduction and importance; Standards setting initiatives and guidelines: COPE, WAME; Conflicts of interest. Publication misconduct: definition, concept, unethical behaviour; Violation of publication ethics, authorship and contributor ship; Identification, complaints and appeals; Predatory publisher and journals. PRACTICE: Plagiarism checking and rectification.

Module-4 – DESIGN OF EXPERIMENTS AND OPTIMIZATION METHODS

16 Hours

Design of experiments; Basic experimental designs; Completely randomized design; Randomized block design; Latin square design; Full factorial design - 2^2 , 2^3 and 2^k ; Fractional factorial designs; Taguchi methodology – Orthogonal array; S/N ratio – Taguchi method for optimization; Response surface method – multi response optimization. Global optimization techniques – evolutionary and swarm algorithms – examples and case studies. PRACTICE: Design of experiment – Taguchi method – Response surface method.

Module-5 – DATA ANALYSIS

17 Hours

Basic probability distributions – Normal distribution – Binominal and Poisson distributions – Weibull and exponential distributions; Sampling: types - size, Sampling tests: T-test F-test – χ^2 test; Modelling – validation and verification; Simulation – techniques; ANOVA; Correlation analyses: Pearson and Spearman Correlations; Regression, classification and cluster analyses; Factor analysis; Discriminant analysis; Uncertainty and error analyses; Machine learning: supervised and unsupervised – examples and case studies. PRACTICE: Distribution; Sampling; Machine learning tools.

Module-1

RESEARCH PREPARATION, PLANNING AND RESOURCES



Research Methodology

- **Research Methodology** is the systematic framework or set of principles that guides how research is conducted. It explains the **overall approach, methods, and processes** used to collect, analyze, and interpret data in order to answer a research question or solve a problem.
 - In simple terms, it is the "**roadmap**" of a research project — describing *what* you are going to do, *how* you are going to do it, and *why* you chose that particular way.
 - It typically includes:
 - **Research design** (qualitative, quantitative, or mixed)
 - **Data collection methods** (surveys, experiments, interviews, observations, etc.)
 - **Sampling techniques**
 - **Data analysis procedures**
 - **Justification** for the chosen approach suitable for reports.
- 



Objectives of Research:

1. To Gain Familiarity with a Phenomenon or to Achieve New Insights (Exploratory Research)

This objective is typically pursued when the researcher is trying to understand a new problem or gain new insights into a situation.

Conducted to explore an area where little is known, identify patterns, or generate new ideas.

Examples:

- Reviewing past studies to identify gaps in electric vehicle charging infrastructure research.
- Conducting focus groups to explore customer expectations for a new mobile banking app.

2. To Portray Accurately the Characteristics of a Particular Individual, Situation or a Group (Descriptive Research)

Descriptive research is conducted to describe systematically a situation, problem, phenomenon, or service.

Aims to describe characteristics, behaviors, or trends without manipulating variables.

Examples:

- Surveying college students to determine their preferred online learning platform.
- Mapping demographic data to describe population distribution in urban areas.
- Tracking annual rainfall patterns in a specific region over the past 20 years.



Objectives of Research:

3. To Determine the Frequency with Which Something Occurs or with Which it is Associated with Something Else (Diagnostic Research)

This type of research focuses on the frequency and relationships among variables.

Focuses on identifying causes of a problem and understanding relationships between variables.

Examples:

- Investigating why a manufacturing plant has high defect rates.
- Analyzing customer complaints to identify root causes of product returns.
- Studying reasons for the decline in student performance in a particular school.

4.To Test a Hypothesis of a Causal Relationship between Variables (Hypothesis-Testing Research)

This objective is pursued when a researcher aims to examine cause-and-effect relationships by testing hypotheses.

Seeks to test a proposed explanation or theory using statistical methods.

Examples:

- Testing whether a new fertilizer increases crop yield compared to the standard one
- Analyzing if a marketing campaign leads to higher sales in target regions.
- Evaluating whether exercise frequency affects blood pressure levels.



Knowledge Economy :

A Knowledge Economy is an economy in which growth is primarily driven by the production, distribution, and use of knowledge and information, rather than the traditional means of production (labor and capital).

Key Features:

Emphasis on intellectual capital and innovation.

Dominance of information and communication technologies (ICTs).

High investment in R&D, education, and skills development.

Examples:

Use of AI in manufacturing.

Data-driven business decisions.

Universities and research labs generating innovations.

Importance in Research Methodology:

Research drives the knowledge economy by producing new knowledge.

Encourages interdisciplinary collaboration.

Focuses on problem-solving, innovation, and technology transfer.

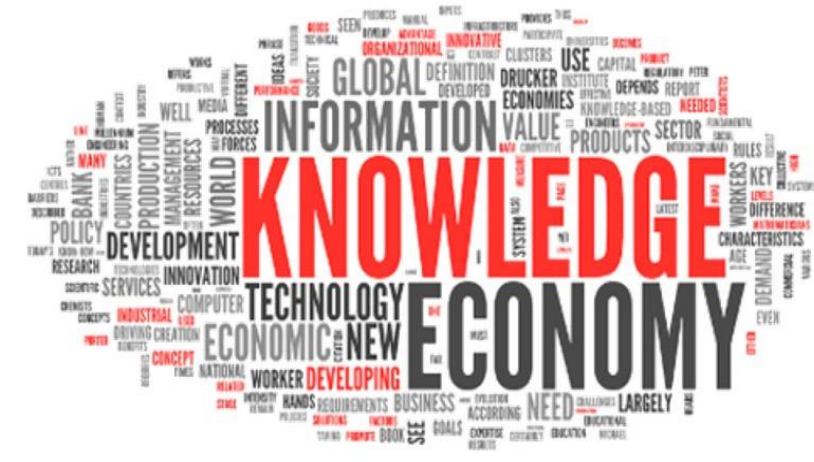


Knowledge Economy :

- **Knowledge Economy** refers to an economic system where **knowledge, information, and intellectual capabilities** are the primary drivers of growth, innovation, and competitiveness—rather than physical resources or manual labor.
- It emphasizes:
 - **Creation** of new knowledge
 - **Dissemination** of information
 - **Application** of expertise to improve products, services, and processes

Examples:

- IT and software development industries
- Research and development sectors
- Biotechnology and pharmaceutical innovations
- Online education and digital content creation



How Knowledge Impacts the Economy:

1.Enhances Productivity:

Knowledge leads to better techniques, automation, and innovation, boosting output per unit of input.

2. Drives Innovation:

Research and development (R&D) create new products, services, and markets.

3. Creates Competitive Advantage:

Countries/companies with better knowledge capital dominate in global markets.

4. Generates Employment in New Sectors:

Growth of sectors like IT, biotech, data science, and clean energy relies on knowledge workers.

5. Promotes Sustainable Development:

Knowledge helps optimize resource use and reduce environmental impact.

6. Improves Decision-Making:

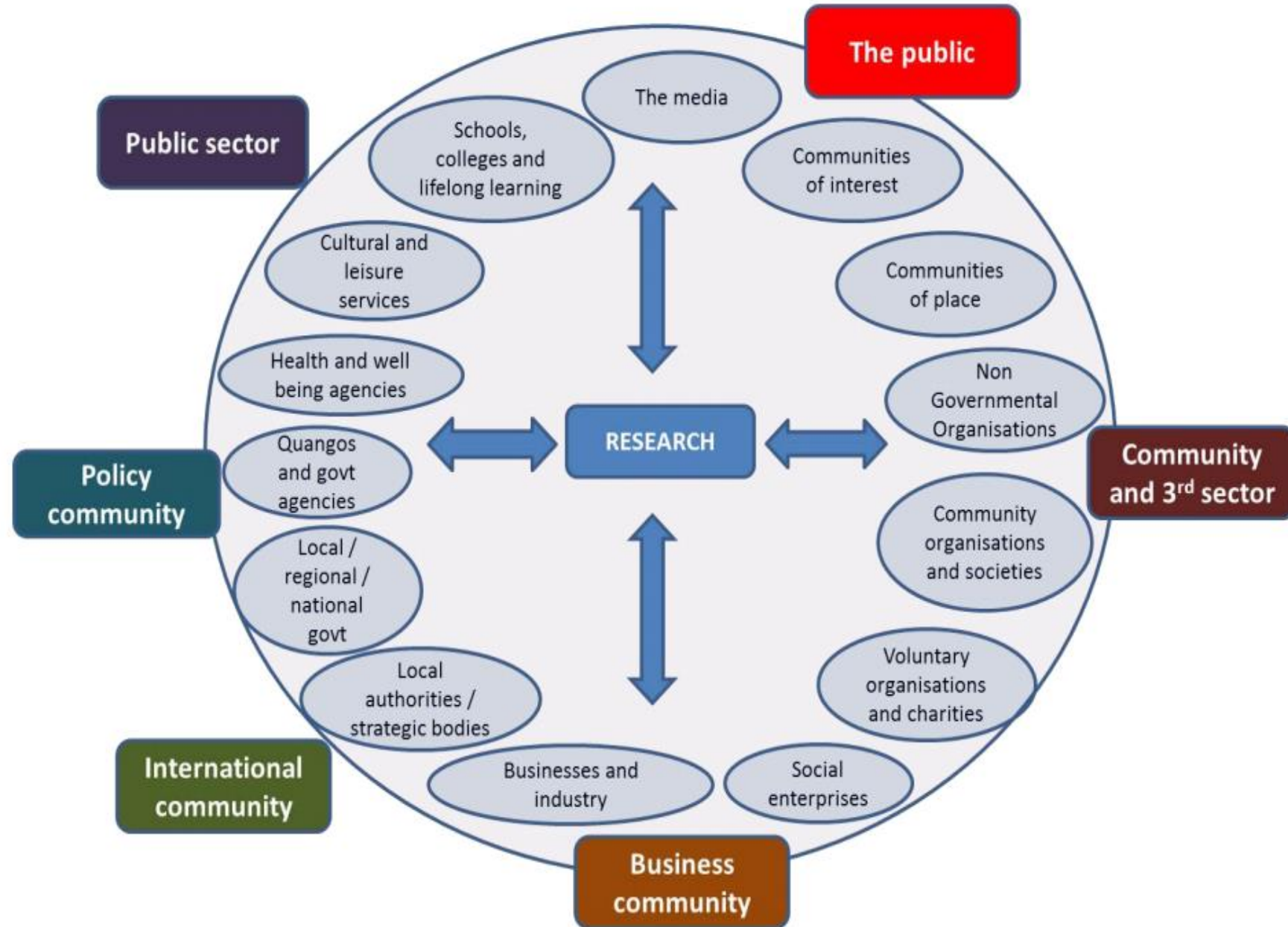
Evidence-based policymaking becomes possible through informed research and data analytics.

Perspectives of Stakeholders in Research

Stakeholders are individuals or groups who are directly or indirectly affected by or involved in the research process.

Key Stakeholder Groups and Their Perspectives:

Stakeholder	Perspective/Interest
Researchers	Aim for academic contribution, discovery, and knowledge creation.
Funding Agencies	Expect useful outcomes, innovation, societal impact, and value for money.
Government/Polycymakers	Use research to shape public policy and economic development.
Industries/Corporates	Expect solutions to real-world problems, innovation, and product development.
Society/Communities	Seek improvement in quality of life, ethical research, and social development.
Academic Institutions	Focus on advancing education, research culture, and publication.



Types of Research

1. Basic (Pure) Research

Basic or pure research is conducted to improve the understanding of fundamental principles and theories. It is not directed toward solving any immediate practical problem but rather focuses on expanding scientific knowledge. For instance, exploring the structure of atoms, understanding the behavior of subatomic particles, or studying genetic sequences fall under basic research. This type of research lays the groundwork for future applied innovations and is driven by curiosity and the desire to uncover truths.

•Purpose:

- Understand natural phenomena or underlying mechanisms.
- Lay the groundwork for **future applied research** and innovations.

•Characteristics:

- Driven by curiosity and intellectual interest.
- Usually theoretical in nature.
- Focused on “**Why**” and “**How**” questions.

•Examples:

- Exploring the **structure of atoms** and subatomic particles in physics.
- Studying **black holes** or the origin of the universe in astronomy.
- Mapping the **human genome** to understand hereditary traits.
- Investigating the **mathematical modeling of fluid dynamics** without current industrial use.

•Significance:

- Leads to groundbreaking discoveries, like the **theory of relativity** or **quantum mechanics**, which later influence technology and industry.

2. Applied Research

Applied research is focused on solving specific, practical problems using existing theories and knowledge. Unlike basic research, it has a clear commercial or societal application. For example, using machine learning models to predict hydrogen storage performance in magnesium-based alloys, or developing a new drug based on clinical trials, are all examples of applied research. It bridges the gap between theoretical knowledge and practical implementation, directly contributing to innovation and development.

- Purpose:**

- Improve existing processes, materials, or technologies.
- Develop new products or applications that directly **benefit society or industries**.

- Characteristics:**

- Practical and solution-oriented.
- Often involves prototypes, field testing, or clinical trials.
- Focused on “**How to**” questions.

- Examples:**

- Developing **hydrogen storage systems** for clean energy using magnesium alloys.
- Creating **AI-based medical imaging software** for early cancer detection.
- Improving **battery efficiency** in electric vehicles.
- Designing **vaccines or drugs** for emerging diseases.

- Significance:**

- Converts basic research findings into **technologies, products, and services**.
- Bridges the gap between theory and practical use, fostering innovation.

3. Descriptive Research

Descriptive research aims to accurately describe the characteristics of a particular individual, group, situation, or phenomenon. It answers the questions of “what,” “when,” “where,” and “how,” without necessarily investigating the causes. Common tools include surveys, case studies, observational methods, and questionnaires. For instance, a demographic study analyzing the percentage of electric vehicle adoption in different regions would be descriptive in nature.

- Purpose:**

- Provide accurate data for **decision-making and future research**.
- Identify patterns, trends, or relationships in the current state of a subject.

- Characteristics:**

- Focused on “**What, When, Where, and How**” rather than “Why.”
- Utilizes **quantitative and qualitative** data collection.
- Often the **first step before exploratory or experimental research**.

- Methods/Tools:**

- Surveys, interviews, and questionnaires.
- Case studies and observational methods.
- Statistical analysis of secondary data.

- Examples:**

- Studying the **adoption rate of electric vehicles** in urban vs. rural regions.
- Mapping **climate patterns and rainfall distribution** in a country.
- Conducting **customer satisfaction surveys** for a product.
- Documenting **biodiversity in a forest reserve**.

- Significance:**

- Helps in identifying **existing conditions and trends**.
- Provides a foundation for **analytical or experimental research**.

4. Analytical Research

Analytical research goes a step beyond description by using existing data or information and analyzing it to make critical evaluations or interpretations. It is commonly used to find patterns, relationships, or trends. Researchers often apply mathematical or statistical tools to understand the data deeply. For example, analyzing the historical performance of hydrogen storage materials and identifying the most influential compositional variables using regression analysis is analytical research.

•Purpose:

- To analyze existing datasets or research findings.
- To identify **patterns, correlations, or trends** for better understanding or decision-making.

•Characteristics:

- Heavily relies on **data analysis, statistics, and logical reasoning**.
- Often involves **secondary data** like reports, records, or published studies.
- Provides a foundation for **predictive modeling and forecasting**.

•Methods/Tools:

- Statistical tests (ANOVA, t-tests, regression analysis).
- Data mining, correlation, and comparative studies.
- Meta-analysis of published research papers.

•Examples:

- Analyzing the **historical performance of hydrogen storage materials** to identify key compositional factors using regression analysis.
- Studying **COVID-19 data trends** to evaluate infection peaks and mortality patterns.
- Evaluating **student performance data** to determine which factors most impact academic success.

•Significance:

- Helps make informed conclusions and **evidence-based recommendations**.
- Converts raw data into **meaningful insights** for decision-making or further research.

5. Exploratory Research

Exploratory research is undertaken when the researcher has little or no knowledge about the problem or phenomenon. The primary objective is to gain insights and familiarity to define problems more clearly and set priorities for future research. This type of research is flexible and open-ended in nature. For instance, if a new type of alloy exhibits unexpected hydrogen behavior, initial lab studies and literature reviews would be exploratory to understand the material better.

•Purpose:

- Understand the **nature of an unknown problem**.
- Generate hypotheses and guide **future detailed studies**.

•Characteristics:

- Flexible, open-ended, and **qualitative in nature**.
- Involves brainstorming, literature review, and pilot studies.
- **Does not provide final answers** but sets the stage for further research.

•Methods/Tools:

- Literature surveys and expert interviews.
- Focus groups and pilot experiments.
- Observational and case study methods.

•Examples:

- Investigating a **new alloy** that exhibits unexpected hydrogen behavior through preliminary experiments and literature review.
- Conducting **pilot studies** on consumer perception of a new electric vehicle design.
- Observing **new social media trends** to understand emerging online behaviors.

•Significance:

- Helps **formulate research problems, hypotheses, or strategies** for deeper investigation.
- Reduces the **risk of misdirection** in subsequent studies.

6. Explanatory Research

Explanatory research, also known as causal research, seeks to identify and understand the reasons behind a phenomenon. It addresses the “why” and “how” questions and aims to explain cause-effect relationships between variables. For example, studying how variations in pressure and temperature affect the hydrogen absorption time of a metal hydride is an explanatory research activity.

- Purpose:**

- Understand **why** and **how** events or behaviors occur.
- Establish relationships between **independent and dependent variables**.

- Characteristics:**

- Quantitative in nature with **controlled experiments or simulations**.
- Often uses **hypothesis testing** to verify causal relationships.
- Requires precise measurements and often **experimental setups**.

- Methods/Tools:**

- Laboratory experiments and simulations.
- Controlled field studies.
- Statistical modeling of causal relationships.

- Examples:**

- Studying how **variations in pressure and temperature** affect hydrogen absorption time in metal hydrides.
- Determining whether **workplace training programs** improve employee productivity.
- Examining the **impact of fertilizer composition** on crop yield under different conditions.

- Significance:**

- Provides **deep understanding** of underlying mechanisms.
- Forms the basis for **prediction, control, and policy-making**.

7. Experimental Research

Experimental research is a scientific approach where one or more variables are manipulated to observe the effect on other variables under controlled conditions. It is commonly used in physical and social sciences, especially in lab settings. This research follows a standard process: hypothesis formulation, designing the experiment, data collection, and statistical analysis. For instance, experimenting with different heat treatment methods on Mg alloys and measuring their hydrogen capacity is experimental research.

•Purpose:

- Establish **cause-and-effect relationships**.
- Test hypotheses in a controlled and repeatable environment.

•Characteristics:

- Conducted in **labs or controlled field environments**.
- Follows a **systematic procedure**:
 1. Formulate a hypothesis.
 2. Design and set up experiments.
 3. Collect and analyze data.
 4. Draw conclusions and validate hypotheses.
- Requires careful **control of external variables** to avoid bias.

•Methods/Tools:

- Laboratory experiments and simulations.
- Randomized Controlled Trials (RCTs).
- Instrumentation for precise measurement (e.g., sensors, testing rigs).

•Examples:

- Experimenting with **different heat treatment methods** on magnesium alloys to measure their hydrogen storage capacity.
- Testing the **effect of varying pressure and temperature** on metal hydride absorption rates.
- Assessing the **impact of a new teaching method** on student performance in a classroom.

•Significance:

- Provides **highly reliable results** due to controlled conditions.
- Essential for developing **new materials, medical treatments, and engineering innovations**.

8. Quantitative Research

Quantitative research focuses on collecting and analyzing numerical data. It uses statistical methods to test hypotheses or predict outcomes and is often used for large-scale studies. Common techniques include surveys with closed-ended questions, experiments, and numerical simulations. An example would be using machine learning algorithms to predict hydrogen desorption times based on alloy composition data.

•Purpose:

- Measure and **quantify variables** for objective evaluation.
- Predict outcomes or trends using **statistical and computational methods**.

•Characteristics:

- Objective and data-driven.
- Uses **large sample sizes** for reliability and reproducibility.
- Employs **mathematical models, statistical tests, and charts** to interpret results.

•Methods/Tools:

- Surveys with **closed-ended questions**.
- Experiments with measurable outcomes.
- Computational simulations and predictive modeling.
- Statistical tools like SPSS, R, Python, MATLAB.

•Examples:

- Using **machine learning algorithms** to predict hydrogen desorption times based on alloy compositions.
- Conducting a **survey of 1,000 electric vehicle owners** to analyze usage patterns.
- Measuring **stress-strain behavior** of composite materials under different loads.

•Significance:

- Supports **data-driven decision-making**.
- Enables **trend forecasting and hypothesis testing** in engineering, healthcare, and social sciences.

9. Qualitative Research

Qualitative research aims to explore deeper meanings, motivations, and experiences. It deals with non-numerical data such as interviews, open-ended surveys, focus group discussions, and content analysis. This type of research is commonly used in behavioral sciences, humanities, and social sciences. For example, understanding how researchers perceive the challenges of adopting AI tools in materials research is a qualitative inquiry.

- Purpose:**

- Gain **in-depth understanding** of human behavior, perceptions, or complex phenomena.
- Identify **patterns, themes, and narratives** rather than numerical trends.

- Characteristics:**

- Subjective and **interpretive in nature**.
- Usually involves **small sample sizes** but provides rich insights.
- Data collected is **textual, visual, or verbal**, not numeric.

- Methods/Tools:**

- Interviews (structured, semi-structured, or open-ended).
- Focus group discussions and ethnographic studies.
- Content analysis of reports, documents, or social media.
- Tools: NVivo, ATLAS.ti for thematic analysis.

- Examples:**

- Understanding **researchers' perceptions of AI tools** in materials science through interviews.
- Exploring **patients' experiences with telemedicine** during the COVID-19 pandemic.
- Studying **cultural influences** on the adoption of renewable energy technologies.

- Significance:**

- Provides **context, meaning, and depth** to complement quantitative studies.
- Useful in **problem exploration, social sciences, and behavioral research**.

10. Action Research

Action research is conducted to find solutions to immediate problems faced by individuals or organizations. It is collaborative and participatory, often involving the people affected by the problem in the research process. Action research is cyclical, involving planning, acting, observing, and reflecting. For example, a research team working with a hydrogen energy lab to improve safety protocols through iterative feedback and improvement cycles is conducting action research.

•Purpose:

- Improve **practices, processes, or performance** in real-world settings.
- Involve stakeholders in the **research and solution-development process**.

•Characteristics:

- Collaborative and participatory**, often involving practitioners and researchers together.
- Cyclical process**: Planning → Acting → Observing → Reflecting → Revising.
- Focuses on **real-time application and improvement** rather than theoretical outcomes.

•Methods/Tools:

- Workshops, focus groups, and feedback sessions.
- Iterative trials and field observations.
- Reflective journals and participatory evaluation.

•Examples:

- A research team **collaborating with a hydrogen energy lab** to improve **safety protocols** through iterative feedback and redesign.
- Teachers **testing and refining new teaching strategies** in classrooms based on student performance feedback.
- Engineers **working with local industries** to reduce waste by continuously modifying processes.

•Significance:

- Provides **practical, immediate solutions** to problems.
- Encourages **stakeholder engagement** and ensures real-world applicability.

11. Cross-sectional Research

Cross-sectional research involves observing a sample or phenomenon at a single point in time. It is often used to determine prevalence or relationships between variables. For example, conducting a one-time survey of researchers' opinions on hydrogen storage materials in 2025 is a cross-sectional study.

•Purpose:

- Provide a **snapshot** of current conditions.
- Identify correlations and patterns **without studying changes over time**.

•Characteristics:

- Usually **quantitative**, though qualitative snapshots are also possible.
- **Cost-effective and time-efficient** since data is collected once.
- Cannot establish **cause-and-effect relationships** due to its single-time nature.

•Methods/Tools:

- Surveys and questionnaires.
- Observational studies and one-time experiments.
- Data analysis of existing reports or databases.

•Examples:

- Conducting a **survey in 2025** to understand researchers' **opinions on hydrogen storage materials**.
- Studying **current battery adoption rates** in electric vehicles in urban areas.
- Evaluating **employee satisfaction** in an organization during a single audit.

•Significance:

- Provides **quick insights** into trends and prevalence.
- Often used in **public health, market research, and social studies**.

12. Longitudinal Research

Longitudinal research studies a subject or sample over an extended period. It helps track changes, patterns, or trends over time. This type of research is especially useful in understanding cause-and-effect relationships. An example would be monitoring the cyclic stability of a hydrogen storage alloy over 5 years to understand degradation behavior.

•Purpose:

- Track **growth, progress, or degradation** over time.
- Identify **long-term effects, causal patterns, and predictions**.

•Characteristics:

- Time-intensive and resource-demanding**.
- Provides **highly reliable data** on progression and trends.
- Useful for **trend analysis and causality studies**.

•Types:

1.**Panel Study:** Observes the **same individuals** repeatedly.

2.**Cohort Study:** Follows a **group with a shared characteristic** (e.g., age group, research field).

3.**Trend Study:** Observes **different samples** over time to identify general trends.

•Methods/Tools:

- Repeated surveys, experiments, and field monitoring.
- Time-series analysis and data visualization.
- Experimental tracking using IoT sensors or lab instrumentation.

•Examples:

- Monitoring the **cyclic stability and degradation** of a hydrogen storage alloy over **5 years**.
- Conducting **long-term clinical trials** to assess the effect of a new vaccine.
- Observing **consumer behavior patterns** in renewable energy adoption over a decade.

•Significance:

- Provides **strong evidence for cause-and-effect relationships**.
- Essential for **policy planning, technological reliability studies, and scientific forecasting**.

Critical and Creative Thinking

□ Critical Thinking

Critical thinking refers to the systematic process of analyzing, evaluating, and interpreting information to form a well-reasoned judgment.

It is objective, logical, and evidence-based, enabling researchers to identify flaws in arguments, spot biases, and question assumptions.

Critical thinkers are skeptical (doubtful) but open-minded, relying on reasoning and data rather than opinions. In research, critical thinking is crucial for evaluating literature, designing methodology, interpreting results, and drawing valid conclusions. For example, critically reviewing conflicting studies on hydrogen storage allows a researcher to assess the strength of evidence, spot gaps, and propose better approaches.

•Purpose in Research:

- Evaluate the **validity, reliability, and relevance** of information.
- Identify **flaws, gaps, and biases** in studies or arguments.
- Make informed **conclusions based on evidence rather than assumptions**.

•Characteristics of Critical Thinkers:

- 1.**Analytical** – Breaks complex problems into smaller parts.
- 2.**Objective** – Avoids personal bias or emotional influence.
- 3.**Logical and Systematic** – Follows clear reasoning steps.
- 4.**Skeptical but Open-minded** – Questions claims but considers new evidence.
- 5.**Evidence-Oriented** – Requires data and facts to support conclusions.

•Applications in Research:

- Literature Review:** Assessing the credibility of existing studies.
- Methodology Design:** Choosing the most valid and reliable approach.
- Data Interpretation:** Avoiding misinterpretation or overgeneralization of results.
- Drawing Conclusions:** Ensuring inferences are **supported by evidence**.

•Examples:

- Critically reviewing **conflicting studies** on hydrogen storage to identify **which experiments are most reliable**.
- Detecting **statistical bias or misinterpretation** in a paper on composite materials.
- Evaluating **the credibility of AI predictions** for material performance before practical application.

•Significance:

- Ensures **accuracy, validity, and credibility** in research outcomes.
- Helps researchers **avoid errors, false assumptions, and weak arguments**.

❑ Creative Thinking

Creative thinking is the ability to look at problems or situations from a fresh perspective and to propose innovative solutions.

It often involves thinking “outside the box,” using imagination, and combining ideas in new ways.

In research, creativity helps in developing novel hypotheses, designing experiments, or using unconventional methods.

A creative researcher might combine techniques from multiple fields (e.g., AI with materials science) to develop new models or technologies. While critical thinking ensures rigor and validity, creative thinking drives originality and discovery. Both are complementary and essential for high-quality research.

•Purpose in Research:

- Generate **novel hypotheses** and innovative solutions.
- Develop **unique experimental designs or interdisciplinary approaches**.
- Overcome research challenges by **thinking beyond conventional methods**.

•Characteristics of Creative Thinkers:

- 1.**Innovative** – Proposes original solutions or perspectives.
- 2.**Curious and Open-Minded** – Explores diverse ideas and approaches.
- 3.**Flexible** – Adapts and modifies approaches when needed.
- 4.**Risk-Taking** – Willing to try unconventional methods.
- 5.**Integrative** – Connects concepts across disciplines.

•Applications in Research:

- Designing **novel experimental setups** for new materials testing.
- Combining **AI, machine learning, and materials science** to create predictive models.
- Developing **new data visualization techniques** to identify hidden trends.

•Examples:

- Integrating **nanotechnology with hydrogen storage research** to develop high-capacity alloys.
- Proposing an **interdisciplinary research method** combining mechanical testing with AI-driven simulations.
- Creating a **new hybrid experimental setup** for faster hydrogen desorption testing.

•Significance:

- Drives **innovation, originality, and discovery** in research.
- Helps overcome **complex, open-ended, or poorly defined problems**.



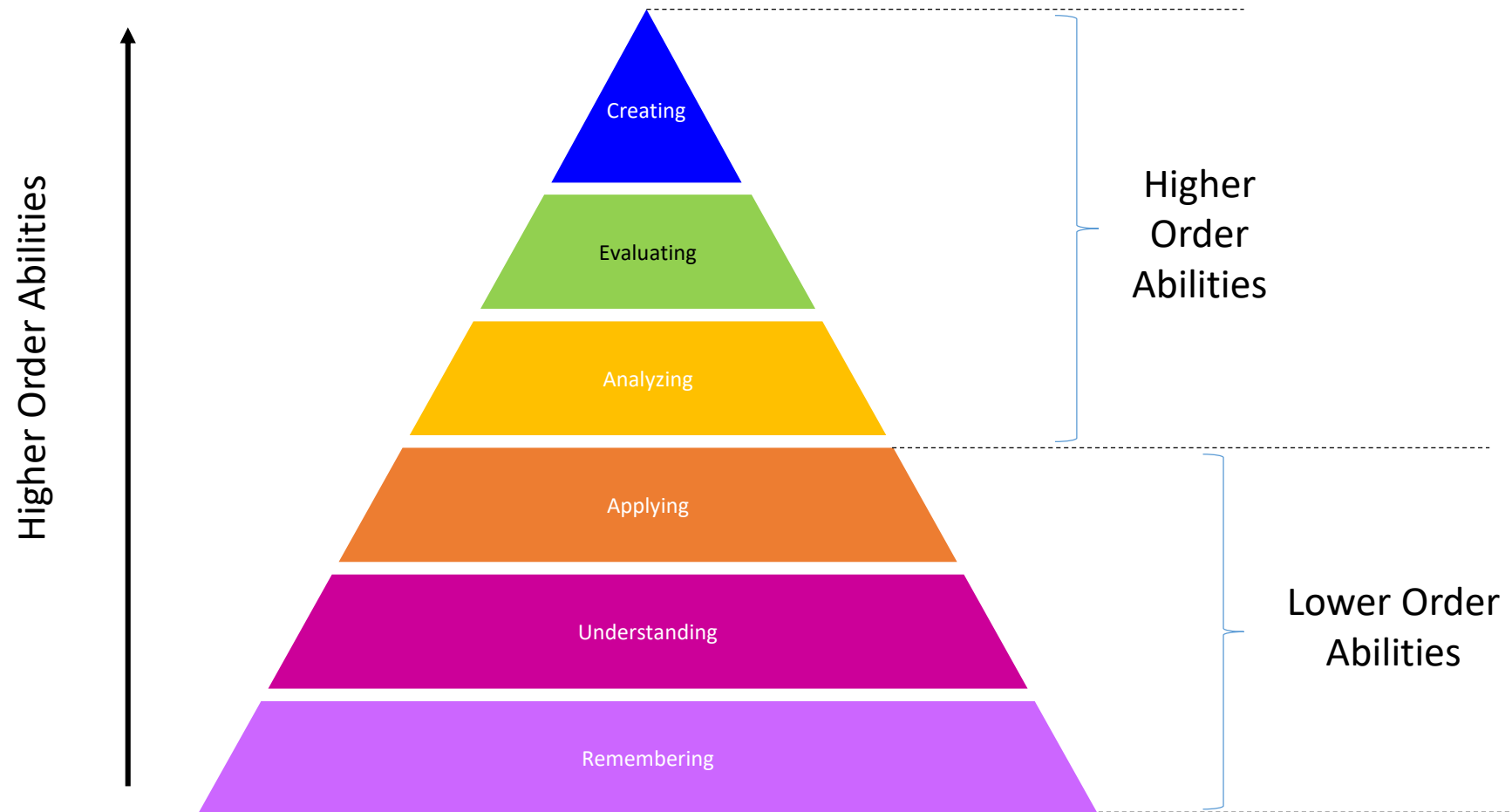
Aspect	Critical Thinking	Creative Thinking
Definition	The process of analyzing, evaluating, and judging information logically to reach a reasoned conclusion.	The process of generating new, original, and innovative ideas or solutions.
Focus	Accuracy, clarity, logic, evidence, and rationality.	Imagination, originality, novelty, and flexibility.
Goal	To make the best possible decision or judgment.	To think of new possibilities and innovative solutions.
Approach	Analytical, convergent (narrowing down to the best option).	Divergent (expanding thinking in many directions).
Skills Involved	Reasoning, questioning, comparing, evaluating, problem-solving.	Brainstorming, imagining, experimenting, exploring alternatives.
Nature of Thinking	Disciplined, structured, evidence-based.	Free-flowing, intuitive, open-ended.
Outcome	A well-supported conclusion or decision.	Unique ideas, new perspectives, creative products.
Example	Evaluating which renewable energy source is most cost-effective for a region.	Designing a completely new type of renewable energy device.

- **Critical thinking = evaluating existing ideas logically.**
- **Creative thinking = generating new ideas imaginatively.**

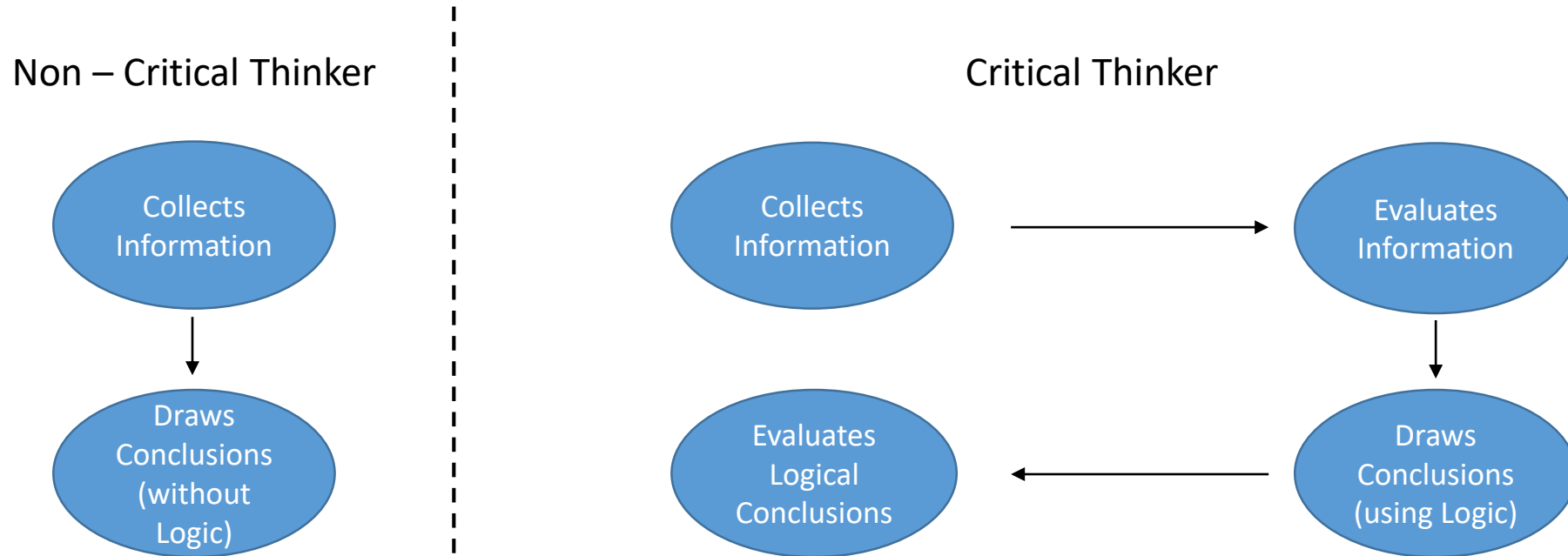
Critical Thinking and Bloom's Taxonomy

- **Critical thinking** is the ability to clearly and logically evaluate information that is presented to us. So, it is an analytical action.

Bloom's Taxonomy Framework



Tools of Critical Thinking



There are four stages in critical thinking:

- i. Collecting data
- ii. Evaluating it
- iii. Drawing logical conclusions and
- iv. Evaluating the validity of the conclusion

Generating a Research Topic

Generating a research topic is one of the most important and challenging steps in the research process.

A good topic is relevant, original, specific, and feasible within the available time and resources.

The process begins by identifying a broad area of interest, such as renewable energy or healthcare, and then narrowing it down through literature review, expert discussion, or real-world observations.

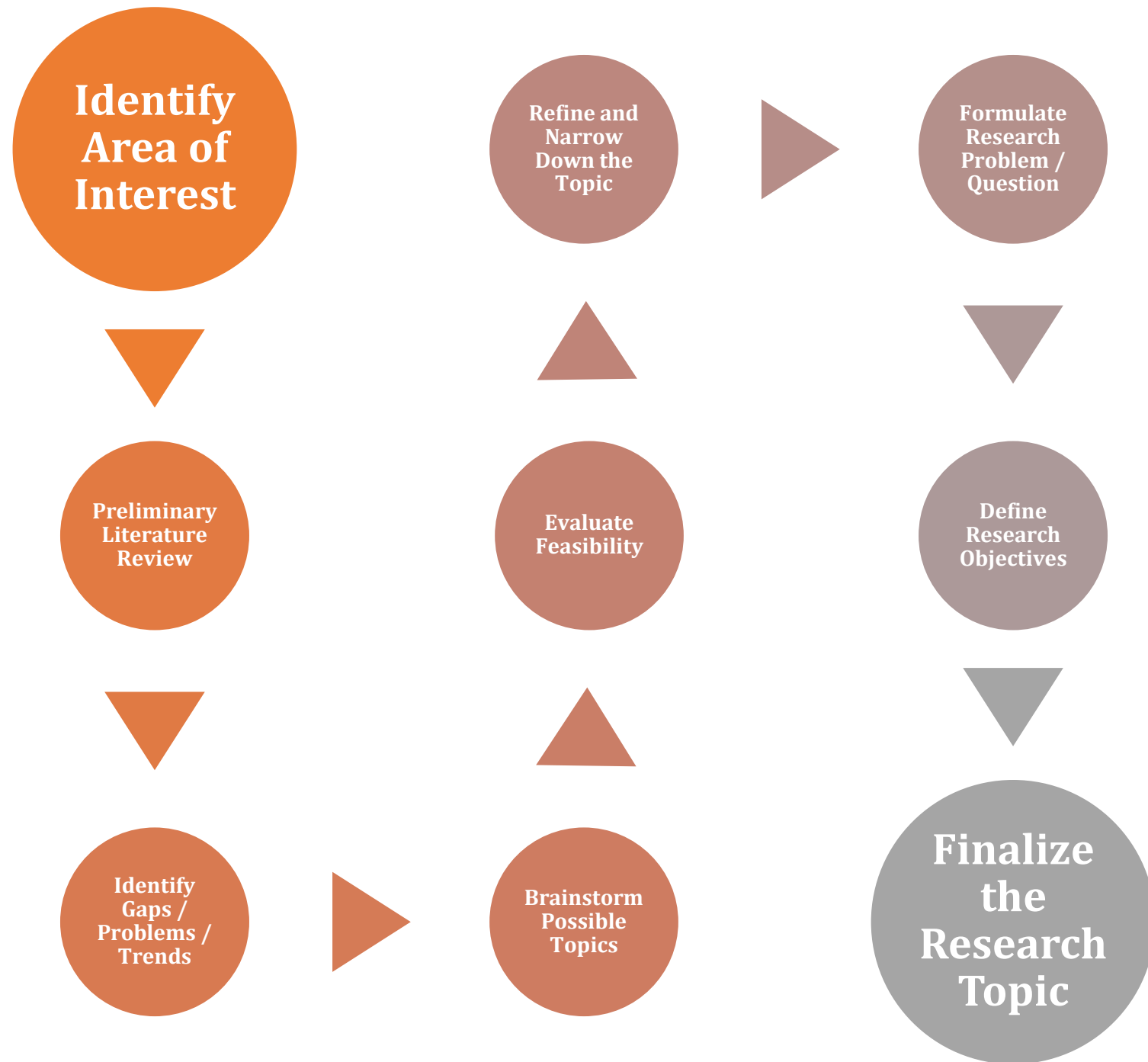
Researchers should explore gaps in existing knowledge, unresolved issues, or areas with practical significance.

Tools like mind mapping, brainstorming, and SWOT analysis can help generate and refine ideas.

Reviewing academic journals, attending seminars, or analyzing trends in the industry can also provide direction.

For instance, if a researcher is interested in magnesium-based hydrogen storage, they may explore under-researched parameters like the role of synthesis pressure on desorption time, eventually formulating a focused, researchable question.

A well-defined research topic leads to clear objectives, guides the research design, and ensures the study adds value to the existing body of knowledge.



Identifying Broad Research Question and Objectives – Design of Questionnaire

□ Identifying Broad Research Question and Objectives

The first and most critical step in any research process is to identify a broad research question that reflects the central theme of the investigation.

A broad research question is an open-ended inquiry that defines the overall direction and scope of the study without getting into specifics at the initial stage. and not too narrow

It typically emerges from a combination of the researcher's interest, preliminary literature review, societal relevance, and knowledge gaps. For example, in the context of hydrogen storage materials, a broad question might be: "What factors influence the hydrogen absorption capacity of magnesium-based alloys?"

Once the broad question is established, the next task is to formulate clear and achievable research objectives.

These objectives break down the broad question into specific, focused goals that guide the methodology and data collection.

Good research objectives should be SMART: Specific, Measurable, Achievable, Relevant, and Time-bound. For instance, one objective could be: "To evaluate the impact of alloying elements on hydrogen desorption temperature in Mg-based alloys."

Objectives act as a roadmap for the entire research process and help in designing appropriate tools for data collection, such as a questionnaire.

Identifying Broad Research Question and Objectives – Design of Questionnaire

□ Identifying Broad Research Question and Objectives

Steps to identify it:

Choose a Research Area – Start with a topic you are interested in (e.g., renewable energy, AI in healthcare, education technology).

Review Literature – Understand what has already been studied and find gaps.

Identify the Problem Context – Frame a large issue (e.g., “How can renewable energy adoption be accelerated in developing countries?”).

Keep It Broad – Don’t focus on variables or metrics yet.

Example BRQs:

“How can electric vehicle charging infrastructure be optimized in urban areas?”

“What are the challenges in integrating AI into healthcare diagnosis?”

“How do digital tools impact student engagement in higher education?”

Identifying Broad Research Question and Objectives – Design of Questionnaire

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Objectives act as a roadmap for the entire research process and help in designing appropriate tools for data collection, such as a questionnaire.

To design and evaluate an intelligent EV charging scheduling algorithm that reduces peak load demand on the grid by at least 20% within 12 months, through the integration of renewable energy sources and real-time load forecasting.”

✓ **Specific** → Focused on intelligent scheduling for EV charging

✓ **Measurable** → 20% reduction in peak load demand

✓ **Achievable** → Feasible with current technology (AI, IoT, renewable integration)

✓ **Relevant** → Directly addresses grid stability and sustainability in EV adoption

✓ **Time-bound** → Achievable within 12 months

Identifying Broad Research Question and Objectives – Design of Questionnaire

□ Identifying Broad Research Question and Objectives

Research Objectives are **specific, measurable goals** that break down the broad research question into smaller, achievable steps.

Types of Objectives:

General Objective – A high-level statement of what you want to achieve.

Specific Objectives – Detailed aims to guide data collection and analysis.

Steps to define objectives:

Start with your broad question.

Decide what you will **explore, describe, or test**.

Use **SMART criteria** (Specific, Measurable, Achievable, Relevant, Time-bound).

Break into **specific sub-objectives** (often 3–5).

Identifying Broad Research Question and Objectives – Design of Questionnaire

□ Identifying Broad Research Question and Objectives

Example:

If BRQ is: *“How can electric vehicle charging infrastructure be optimized in urban areas?”*

General Objective:

To explore strategies for improving electric vehicle charging infrastructure in cities.

Specific Objectives:

To analyze the current distribution of EV charging stations.

To identify user needs and charging patterns.

To evaluate the role of government policies in infrastructure development.

To propose a model for optimal station placement.

1. Broad Research Question

•Importance:

- Sets the foundation of the study.
- Helps focus on relevant knowledge gaps or societal needs.

•**Sources:** Literature review, industry problems, knowledge gaps, personal interest.

•Characteristics:

- Open-ended, relevant, and researchable.
- Provides scope for refinement and exploration.

•Examples:

- *“What factors influence hydrogen absorption in Mg-based alloys?”*
- *“How does urbanization affect local biodiversity?”*

2. Research Objectives

•**SMART:** Specific, Measurable, Achievable, Relevant, Time-bound.

•Importance:

- Guide methodology and data collection.
- Provide clarity and measurable outcomes.

•Examples:

- *To evaluate the effect of alloying elements on hydrogen desorption temperature.*
- *To analyze the role of microstructure in hydrogen absorption kinetics.*

Aspect	Broad Research Question	Objectives
Nature	Open-ended, exploratory	Focused, actionable
Scope	Wide	Narrow, specific
Purpose	Sets the direction of study	Guides data collection & analysis
Example	“How does AI impact medical diagnosis?”	“To assess the accuracy of AI vs. doctors”

3. Linking Questions to Objectives

• **Process: Broad Question:** *“What factors influence hydrogen absorption in Mg alloys?”*

- **Objectives:**
 - *Investigate the effect of microstructure on absorption capacity.*
 - *Study the impact of alloying elements on desorption temperature.*

□ Design of Questionnaire

Once the research objectives are clear, the next step is to design a questionnaire to collect primary data aligned with those objectives. A questionnaire is a structured tool that consists of a series of questions used to gather information from respondents. It is widely used in survey-based research and should be carefully crafted to ensure reliability and validity of the data.

The design process begins with identifying the type of data needed – qualitative, quantitative, or both – based on the research objectives. Then, suitable question types are selected, such as open-ended, close-ended, Likert scales, ranking, or multiple-choice questions. Each question should be clear, unbiased, and relevant to a specific research objective. The questionnaire should be logically ordered, starting with general questions and gradually moving to specific ones, to ensure respondent engagement and flow.

It is also important to pilot test the questionnaire with a small group before actual data collection. This helps in identifying ambiguities, improving question wording, and ensuring the questionnaire is easy to understand. In summary, a well-designed questionnaire is a direct reflection of the research objectives, ensuring that the data collected is both accurate and useful for answering the research question.

Based on your experience, share your satisfaction on the following parameters

(1=Very Dissatisfied & 5=Very Satisfied)

	1	2	3	4	5
Ease of placing the order	☐	☐	☐	☐	☐
Speed of delivery	☐	☐	☐	☐	☐
Packaging	☐	☐	☐	☐	☐

•Identify Data Needs

- Decide on **quantitative, qualitative, or both** data types.
- Example:* Measure researchers' views on hydrogen storage methods.

•Select Question Types

- Close-ended:** MCQs, Yes/No, Likert scale → Easy to analyze.
- Open-ended:** Opinions or detailed responses → Rich insights.
- Ranking/Rating:** Prioritize or evaluate factors.

•Ensure Clarity & Flow

- Simple, **unbiased, and objective-linked** questions.
- Logical order: **General → Specific**.

•Pilot Testing

- Test with a small sample to fix **ambiguities or confusing wording**.

•Finalize & Deploy

- Ensure **concise, clear, and well-structured** format.
- Use online tools like **Google Forms** or SurveyMonkey.

Example (Hydrogen Storage)

1. *Which Mg-based alloy do you use for hydrogen storage?* (MCQ)
2. *What is the biggest challenge in Mg alloy hydrogen absorption?* (Open-ended)
3. *Rate the importance of these factors in storage efficiency:*
 1. Microstructure
 2. Alloying elements
 3. Temperature/pressure (Likert 1–5)

Characteristics of a Good Research

A good research study is one that is systematic, logical, and purposeful, and meets academic or scientific standards. It possesses several essential characteristics that ensure the quality, reliability, and usefulness of the findings.

1. Clearly Defined Objectives

Good research begins with well-defined and specific objectives. These objectives guide the direction of the study and help in focusing efforts on relevant aspects of the problem.

2. Systematic Process

The research process follows a structured and orderly method involving steps like problem definition, data collection, analysis, and interpretation. It is not haphazard or arbitrary.

3. Empirical Evidence

A strong research study relies on observable and verifiable data, whether quantitative or qualitative. The results are based on facts gathered through experiments, surveys, or observations.

4. Logical Reasoning

Research involves inductive and deductive reasoning to draw valid conclusions. The research logic should be consistent and support the objectives and hypotheses.

5. Replicability

The methodology used in good research should be described clearly so that other researchers can replicate the study and verify its results.

6. Objectivity and Neutrality

Good research is free from personal biases, preferences, or subjective opinions. The researcher must remain neutral while collecting and analyzing data.

7. Ethical Practices

Ethical conduct, including informed consent, privacy, and honesty in reporting results, is a cornerstone of good research.

8. Usefulness and Applicability

The findings of good research should contribute to knowledge, help solve real-world problems, or provide new insights. The results should be meaningful and applicable to relevant fields.

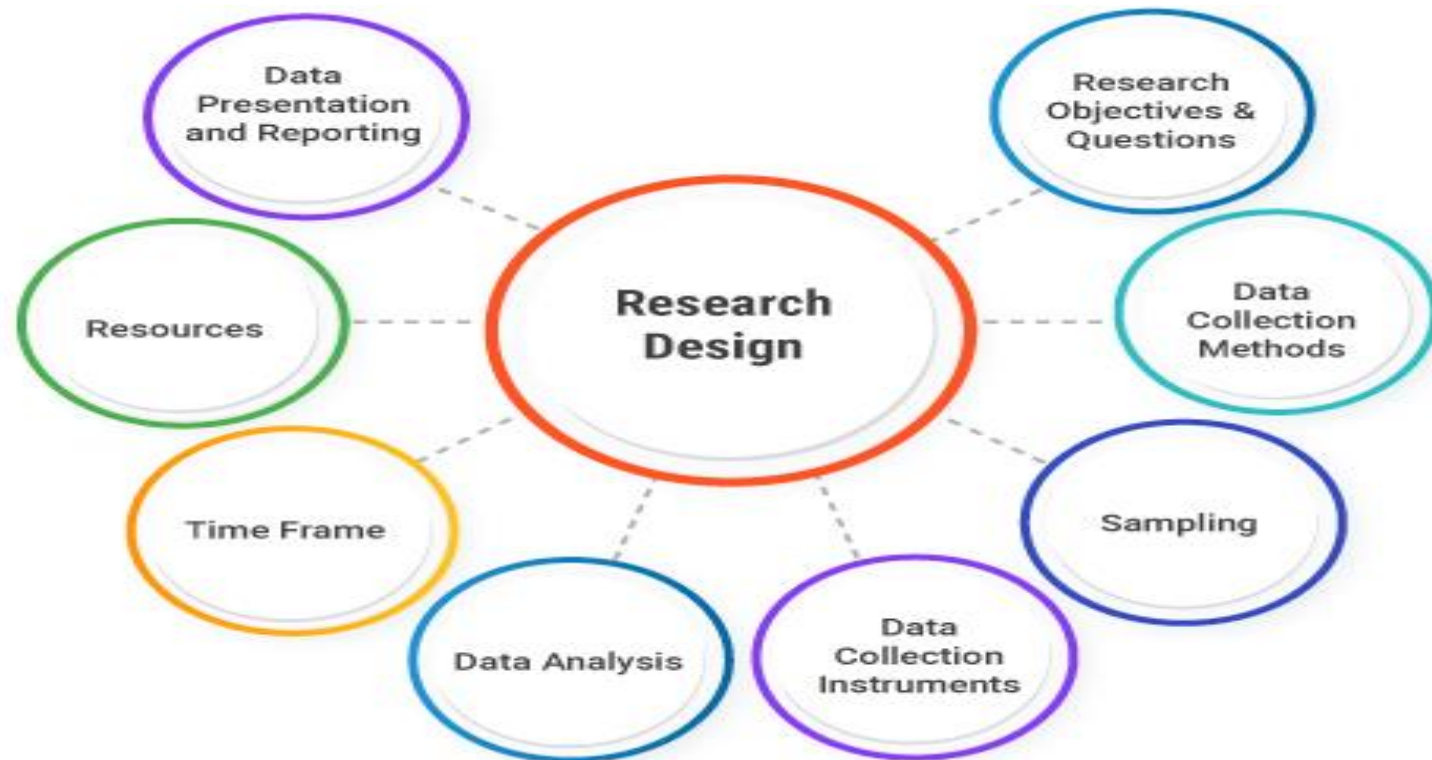
9. Originality and Innovation

Good research adds something new—either through original data, new methods, or fresh interpretations—to the existing body of knowledge.

Research Design

Definition

Research design is the framework or blueprint for conducting a research project. It lays out the procedures for collecting, measuring, and analyzing data in a way that answers the research questions or tests the hypotheses. A good research design ensures that the evidence obtained enables the researcher to effectively address the research problem with accuracy and efficiency.



Key Components of a Research Design

1. Research Problem

The design begins with identifying and clearly stating the research problem or question to be investigated.

2. Objectives of the Study

The design should clearly mention what the research aims to achieve – whether it is to explore, describe, analyze, or predict something.

3. Type of Data Required

Determines whether the research will use primary data (collected directly) or secondary data (from existing sources).

4. Methods of Data Collection

This includes surveys, interviews, observations, experiments, or document analysis. The method must align with the research objectives.

5. Sampling Design

It explains the target population, sampling frame, sample size, and sampling technique (random, stratified, purposive, etc.).

6. Tools and Techniques of Analysis

Specifies the statistical or qualitative methods to be used for analyzing data (e.g., regression, ANOVA, thematic coding).

7. Timeframe and Resources

Outlines how long the research will take, the stages involved, and the resources (funds, personnel) required.

Types of Research Design

Research designs can be classified based on the nature and purpose of the study:

1. Exploratory Research Design

This is used when the problem is not well understood. It aims to explore the issue and generate new ideas. It is flexible and often uses interviews, focus groups, or literature review.

2. Descriptive Research Design

Descriptive design aims to describe characteristics of a population or phenomenon. It answers the "what," "where," and "how" questions but not "why."

3. Diagnostic Research Design

Used to determine the causes of a particular phenomenon or problem. It involves in-depth analysis and often compares groups to identify patterns.

4. Analytical Research Design

Involves the analysis of existing data or information to evaluate hypotheses or draw conclusions. Often used in meta-analyses or modeling studies.

5. Experimental Research Design

Here, the researcher manipulates one or more variables and observes the effects under controlled conditions. It is used to establish cause-effect relationships.

6. Cross-sectional Design

Studies a phenomenon at a single point in time. Useful for surveys and prevalence studies.

7. Longitudinal Design

Involves studying the same subjects over a long period to identify trends or changes.



Methods of Scientific Enquiry

Scientific enquiry refers to the systematic investigation of phenomena through observation, experimentation, and logical reasoning to discover new knowledge or validate existing ones. It is the foundation of research in natural and social sciences, and it follows a structured and objective approach to understand how things work. The major methods used in scientific enquiry include theoretical, experimental, and empirical approaches.

1. Theoretical Method

The theoretical method involves the use of logic, reasoning, mathematical modeling, and abstract thinking to understand and explain phenomena. It is often used when direct observation or experimentation is not possible or practical. Theoretical enquiry helps in developing models, hypotheses, and predictions that can later be tested through empirical or experimental methods. For example, a physicist might use equations and simulations to describe the behavior of particles at the quantum level, or a chemist may model how hydrogen atoms interact with magnesium in storage alloys. Theoretical research provides the framework and foundation for designing experiments and interpreting data. It is essential for building scientific theories and expanding conceptual understanding.

2. Experimental Method

The experimental method is perhaps the most recognized approach in scientific enquiry. It involves deliberately manipulating variables to observe and measure their effects under controlled conditions. This method is used to test hypotheses, establish causal relationships, and validate theoretical predictions.

In a laboratory setting, researchers may control temperature, pressure, or concentration while measuring how these changes impact a dependent variable. For example, in hydrogen storage research, an experiment might be designed to test how different alloy compositions affect absorption time. The strength of the experimental method lies in its accuracy, replicability, and ability to establish cause-effect relationships.

3. Empirical Method

The empirical method is based on direct observation, experience, and measurement. Unlike theoretical methods which rely on abstraction, empirical enquiry relies on real-world data. It involves collecting data through surveys, field studies, interviews, observations, and measurements. It is commonly used in both natural and social sciences.

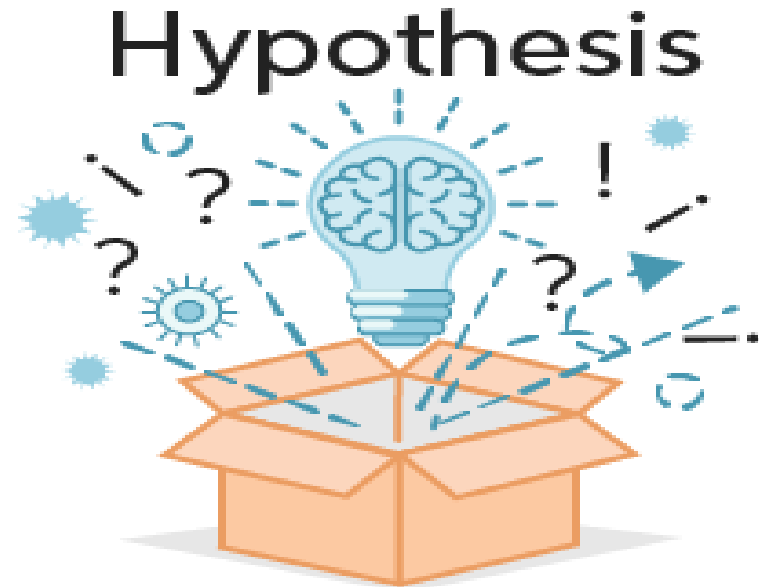
For instance, a materials scientist conducting real-time monitoring of hydrogen release from a sample under operational conditions is using empirical methods. In the social sciences, studying user behavior toward clean energy adoption via surveys also follows the empirical approach. Empirical data is critical in validating theories and making informed decisions.

Hypothesis – Formulation & Testing

What is a Hypothesis?

A hypothesis is a tentative, testable statement or prediction about the relationship between two or more variables. It serves as the foundation for scientific enquiry and guides the direction of research. A hypothesis is essentially an assumption that the researcher tries to prove or disprove through data collection and analysis. For example: “Increasing the magnesium content in alloys decreases hydrogen absorption time.”

A good hypothesis must be clear, specific, testable, and falsifiable. It should be based on existing knowledge and capable of being supported or refuted by empirical evidence.



Hypothesis Formulation

A hypothesis is an assumption made based on some evidence, which is tested through experiments or statistical analysis. It helps in converting a research problem into a testable statement.

Steps in Hypothesis Formulation:

- ☐ Identify the research problem.
- ☐ Define dependent and independent variables.
- ☐ Establish the relationship (positive/negative/no relation).

There are typically two types of hypotheses:

- ☐ **Null Hypothesis (H_0):** States that there is no relationship between the variables.

Example: “There is no effect of temperature on hydrogen desorption rate.”

- ☐ **Alternative Hypothesis (H_1):** States that there is a relationship or effect between the variables.

Example: “Higher temperature increases the hydrogen desorption rate.”

A hypothesis can also be directional (specifying the nature of the relationship) or non-directional (simply stating that a relationship exists without indicating the direction).

Example Problem:

A researcher wants to study whether a new Mg-based alloy reduces hydrogen absorption time compared to the existing alloy.

Population mean absorption time for the existing alloy: $\mu = 4200$ seconds

New sample data ($n = 10$):

3900, 3950, 4000, 4100, 4050, 3990, 3920, 3960, 4010, 4020

Formulated Hypotheses:

Null Hypothesis (H_0): $\mu = 4200$

(The new alloy has no improvement in absorption time.)

Alternative Hypothesis (H_1): $\mu < 4200$

(The new alloy has significantly reduced absorption time.)

This is a left-tailed test (directional).

Significance Level (α) : The significance level, usually denoted by α (alpha), is the threshold set by the researcher to determine how much risk of making a wrong decision is acceptable when testing a hypothesis. It defines the probability of rejecting a true null hypothesis (i.e., making a Type I error).

p-value (Probability Value) : The p-value is the calculated probability of obtaining the observed results (or more extreme) assuming that the null hypothesis is true.

Testing of Hypothesis

Hypothesis testing is a statistical procedure used to evaluate whether the data collected supports the null hypothesis or leads to its rejection in favor of the alternative hypothesis. The steps involved are:

- ❑ Formulate the Hypotheses

Define both the null (H_0) and alternative (H_1) hypotheses based on your research question.

- ❑ Select a Significance Level (α)

Commonly set at 0.05 (5%), this is the probability of rejecting a true null hypothesis (Type I error).

- ❑ Choose the Appropriate Test

Depending on the data type and objective, use tests such as t-test, z-test, chi-square, or ANOVA.

- ❑ Compute the Test Statistic

Calculate the value based on the sample data using statistical formulas or software.

- ❑ Make a Decision

Compare the test statistic with the critical value (from statistical tables) or check the p-value:

- If $p\text{-value} \leq \alpha$, reject the null hypothesis.
- If $p\text{-value} > \alpha$, fail to reject the null hypothesis.

- ❑ Interpret the Results

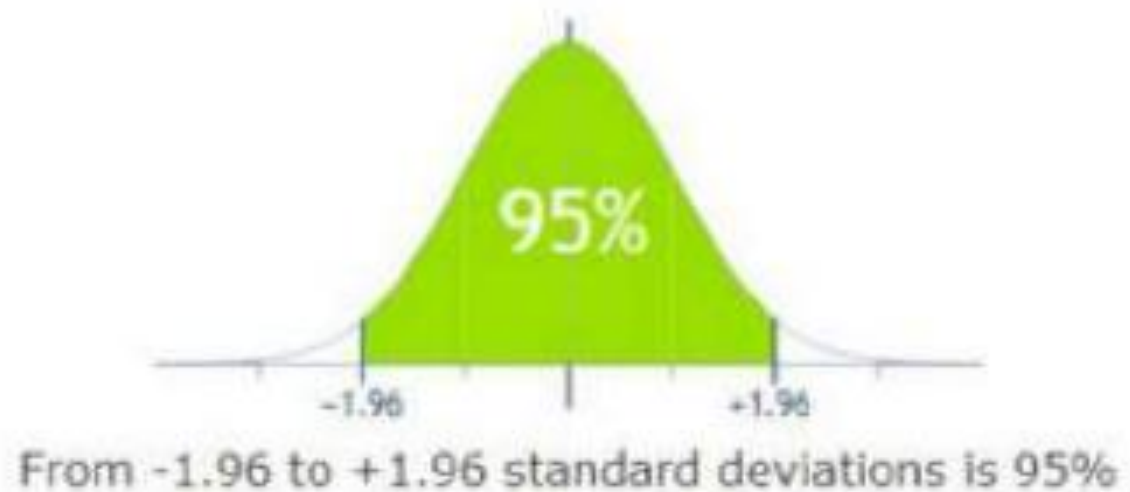
Conclude whether the evidence supports the alternative hypothesis or not.

Hypothesis Testing with Example

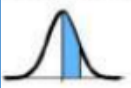
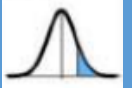
- A manufacturer of switches claims that data obtained over the years, show that their switches lasts for a **mean of 2,000 hours** with a **standard deviation of 200 hours**.
- Later, the R&D department claimed that they have came up with some innovation that might extend the life of the switches.
- To test this, the company conducted a test with **100 switches** which showed that the they now had an mean life of **1,960 hours** with a standard deviation of **180 hours**.
- Based on these results, can the manufacturer claim that this has led to a change in the life span of the switches? Frame a hypothesis and test it with a significance level of 5%.

- **Solution** - Based on the claims:
- **Population Parameters:**
- Population mean, $\mu = 2,000$ hours of switch life
- Population standard deviation, $\sigma = 200$ hours of switch life
- Hypothesis:
- We can propose Null Hypothesis : $H_0 : \mu_0 = 2,000$
- Alternate Hypothesis would then be : $H_1 : \mu \neq \mu_0$, So $\mu > \mu_0$, or $\mu < \mu_0$,
- The significance level as suggested, $\alpha = 0.05$
- **Sample Data :**
- Sample size as provided, $n = 100$
- Sample Mean, $\bar{X} = 1,980$ hours of switch life
- Sample Standard Deviation, $\sigma = 180$
- As $n > 30$, we can assume normal distribution which would make the standard deviation to be, $\sigma = 200$
- As the alternate hypothesis, $H_1 : \mu \neq \mu_0$, So $\mu > \mu_0$, or $\mu < \mu_0$, we can use the two-tailed test for testing the hypothesis.

- The probability (area) under the standard normal distribution corresponding to $Z_{-\alpha/2} = Z_{-0.025}$ and $Z_{-\alpha/2} = Z_{+0.025}$ can be read from the table as shown.
- The values obtained are $Z = +1.96$ for $Z_{+0.025}$ and would be $Z = -1.96$ for $Z_{-0.025}$ due to symmetry of the Normal distribution



Standard Normal
Distribution (Z) Probabilities

		
z	Pr (0 < Z ≤ z)	P(Z ≥ z)
1.79	0.463273	0.36727
1.80	0.464070	0.035930
1.81	0.464852	0.035148
1.82	0.465621	0.034380
1.83	0.466375	0.033625
1.84	0.467116	0.032884
1.85	0.467843	0.032157
1.86	0.468557	0.031443
1.87	0.469258	0.030742
1.88	0.469946	0.030054
1.89	0.470621	0.029379
1.90	0.471283	0.028717
1.91	0.471933	0.028067
1.92	0.472571	0.027429
1.93	0.473197	0.026803
1.94	0.473810	0.026190
1.95	0.474412	0.025588
1.96	0.475002	0.024998
1.97	0.475581	0.024419
1.98	0.476148	0.023852
1.99	0.476705	0.023296
2.00	0.477250	0.022750
2.01	0.477784	0.022216

- The calculated value of Z would be given by the expression
 - $Z = \frac{x - \mu}{\sigma / \sqrt{n}}$
- Where σ is the sample standard deviation (not population standard deviation)
- Substituting $\implies Z = \frac{1960 - 2000}{180 / \sqrt{100}} = -2.22$
- As $-2.22 < -1.96$, the z value falls in the tail region and therefore the null hypothesis is rejected!
- Hence the changes have indeed significantly changed the life of the switches!

Sources of Information – ICT Enabled Tools

Information and Communication Technology (ICT) has revolutionized how researchers access, manage, and disseminate information. ICT-enabled tools provide fast, reliable, and vast sources of knowledge crucial for academic research.

Digital Libraries – Access academic papers and books.

Example: IEEE Xplore, Shodhganga

Online Journals – Latest research articles and reviews.

Example: ScienceDirect, SpringerLink

Search Engines – Find scholarly content quickly.

Example: Google Scholar, Scopus

Reference Managers – Organize and cite sources.

Example: Mendeley, Zotero

Online Surveys – Collect data via forms.

Example: Google Forms, SurveyMonkey

E-learning Platforms – Learn research methods and tools.

Example: NPTEL, Coursera

Academic Networks – Connect with researchers.

Example: ResearchGate, Academia.edu



1. Databases

Organized digital collections that store academic and scientific data.

Example: Scopus, Web of Science, PubMed.

2. Repositories

Digital archives for storing research outputs like theses, reports, datasets.

Example: Shodhganga (India), arXiv, Dryad.

3. Public Sources

Freely available government or institutional data and reports.

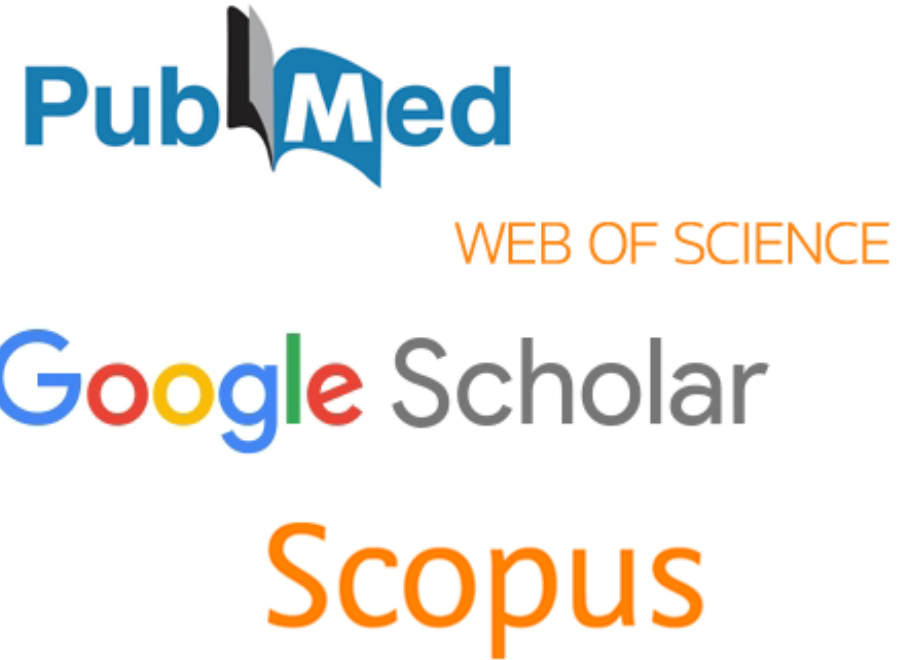
Example: Census data, National Sample Survey (NSS), WHO, World Bank.

4. Private Sources

Commercial or subscription-based information services.

Example: Gartner reports, Elsevier, Bloomberg, Market research firms.

Let me know if you want a compact version for presentation slides.



Indexes

Indexes are structured tools used to systematically organize and locate published literature. In research, an index typically refers to bibliographic databases that allow users to search for articles, papers, journals, and books based on authors, titles, subjects, or keywords. Examples include Scopus, Web of Science, and PubMed, which index thousands of journals and help researchers access relevant publications efficiently. Indexes are essential for tracking citation metrics and finding high-impact journals.

Literature Search – Keyword

A keyword-based literature search is one of the most fundamental methods to identify relevant research papers, articles, or books. It involves selecting appropriate keywords or key phrases that represent the core ideas of the research topic. These keywords are then used in databases or search engines to filter through large volumes of data. For example, for a study on magnesium-based hydrogen storage, keywords might include “magnesium alloy,” “implants,” and “teeth.” Boolean operators like AND, OR, and NOT can refine the search further.

Backward and Forward Search

These are two effective strategies for conducting citation-based literature reviews:

- ❑ **Backward Search:** Also known as reference mining, this involves going through the reference list of a relevant paper to discover earlier works that it cited. This helps in understanding the background and theoretical framework that led to the current study.
- ❑ **Forward Search:** This method tracks who has cited a particular paper after its publication. It helps in identifying newer studies that have built upon the original work. Tools like Google Scholar, Scopus, and Web of Science allow researchers to easily perform forward citation tracking.

Aspect	Backward Research	Forward Research
Definition	Starts with a desired outcome or decision and works backward to determine what information is needed to achieve it.	Starts with curiosity or a research question and works forward through exploration to discover insights.
Approach	Goal-driven, decision-oriented	Exploration-driven, discovery-oriented
Focus	Solve a specific problem or make a decision	Gain new understanding or generate knowledge
Planning	Clear objectives and end-goals first, then define data to collect	Begin with broad questions, refine objectives as you go
Data Use	Collect only data directly needed for the decision	Collect broader data for analysis, trends, and patterns
Best For	Market research, product design decisions, feasibility studies	Academic research, R&D, innovation exploration
Example in EV Industry	Deciding whether to launch a new EV model in a specific market	Exploring new battery chemistries for next-gen EVs

Backward Research Example (Goal → Steps)

•**Goal:** Should we launch a mid-range EV in Tier-2 Indian cities?

•**Process:**

- Start with the **decision needed** (launch or not).
- Identify the **key metrics** (income levels, EV adoption trends, charging infra).
- Collect targeted data: surveys on price sensitivity, dealer availability, charging networks.
- Analyze if Return Of Interest meets company benchmarks.

•**Outcome:** A clear **yes/no recommendation** for launching that EV model.

Forward Research Example (Question → Discovery)

•**Question:** How can we increase EV battery efficiency by 20%?

•**Process:**

- Start with a **broad question** about improving energy density.
- Explore battery chemistries: solid-state, graphene, lithium-sulfur.
- Conduct lab experiments, simulations, long-term performance tests.
- Generate **new knowledge** that might lead to breakthroughs.

•**Outcome:** Insights for **innovations**, patents, or academic publications.

Search in: Web of Science Core Collection ▾

DOCUMENTS

AUTHORS

CITED REFERENCES



Cited Author ▾



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Cited Work ▾



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Example: 2013-2014

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X CLEAR

SEARCH



ScienceDirect



Search for peer-reviewed journals, articles, book chapters and [open access](#) content.

Keywords

Author name

Journal/book title

Volume

Issue

Pages



[Advanced search](#)

Quality Measurement in Research

Evaluating the quality and impact of research is crucial to ensure its validity, relevance, and contribution to the academic community. Quality measurement tools provide quantifiable indicators to assess the influence, credibility, and productivity of research outputs.

Journal Metrics

Journal metrics are tools used to evaluate the influence and reputation of academic journals. They help researchers choose suitable platforms for publication and are often used by institutions and funding agencies for evaluation.

Examples:

Impact Factor (IF): Measures the average number of citations received per paper published in a journal during the preceding two years.

SCImago Journal Rank (SJR): Considers both the number of citations and the prestige of the journals where such citations come from.

h5-index (by Google Scholar): Measures the largest number h such that h articles published in the last five years have at least h citations each.

Eigenfactor Score: Reflects the journal's total importance to the scientific community, based on citation networks.

Research Journal Indexing and Quality Standards

Factors Necessitating Development of Journal Citation Indexing

There are three factors that led to development of citation indexing back in 1950's:

- Followed by World War II, there were huge influx of government funds for research. Publication of research finding by researchers burgeoned the body of knowledge. In order to retrieve from the huge document database, it became necessary to index it. Manual indexing practice is time consuming and not a cost effective one.
- The then method of subject indexing was not useful because of discipline specific terminology etc.
- Availability of machines for computation of data spurred citation indexing
- Citation Indexing proposed and developed by Eugene Garfield.

- When a manuscript is ready for submission, from the **researcher perspective**, the criteria used for selecting the journal is the it's **subject focus**, however, other factors such as **good reputation** with **indexing** in major databases also needs to be considered.
- This subject focus + reputation + indexing ensures:
 - Good potential for exposure of the work to research community
 - Greater chance of being discovered and cited by other researchers
 - Greater potential for garnering citations, thereby be able to track the opinion of other researchers and learn how the work contributed to the science and society.
- The **reputation** of the journal is judged by **quality standards** as assessed by **impact factor** and **source normalized impact factor**.
- From the **research journal's perspective**, the prestige of the journal is determined by the abstracting and indexing services covering it, which in turn is linked to the citations received by articles published in it

Citation Indices

- The first modern citation index was suggested by information scientist **Eugene Garfield** in **1955** and created by him in 1964, and it represents an important innovation to knowledge organization and information retrieval.
- Current important citation indices are:
 - Clarivate Analytics citation indices (**WoS**) - Science Citation Index (SCI); Science Citation Index Expanded (**SCIE**); Social Sciences Citation Index (SSCI); Arts & Humanities Citation Index (A&HCI); Conference Proceedings Citation Index (CPCI); Book Citation Index (BKCI); Data Citation Index Emerging Source Citation Index (**ESCI**)
 - CiteSeerx-since 1997, mainly computer and info. science
 - Crossref-since 2000 by Publishers Association using the DOI
 - Scopus Citation Index (**Scopus**) - since 2004 by Elsevier, largest database
 - Google Scholar (GS)-since 2004 by Google Inc., but poor metrics
 - Microsoft Academic (MA)- since 2016, poor bibliometrics tool
 - Dimensions-since 2018 by Digital Science

Important Citation Indices

Web of Knowledge/Science based by Clarivate Analytics

- Science Citation Index (SCI) and Science Citation Index Expanded (SCIE)- (established 1964, database 1899 to present)
 - A multidisciplinary database, with searchable author abstracts, covering the journal literature of the sciences.
 - It indexes more than 6,650 major journals across 150 scientific disciplines.
- Social Science Citation Index (SSCI) - (established 1973, database 1898 to present)
 - Covering journal literature of the social sciences It indexes more than 1,950 journals spanning 50 disciplines
- Arts & Humanities Citation Index (A&HCI) -(established 1978, database 1975 to present)
 - Covering the journal literature of the arts and humanities
 - It indexes more than 1,395 of the world's leading arts and humanities journals.

Other Important Citation Indices

- Conference Proceedings Citation Index-Science (CPCI-S) - (established in 2008, database from 1991 to present)
 - Lets you use cited reference searching to see the full impact of conferences and other professional meeting
 - With a global coverage of over 110,000 conference proceedings
- Conference proceedings Citation Index-Arts & Humanities (CPCI-A&H)-(established in 2008, database 1991 to present)
 - This citation index covers conference literature in all fields of social sciences, arts and humanities.

Scopus Database based by Elsevier

- Scopus (established in 2004, database mainly 1996 to present)
 - Largest abstract and citation database of peer-reviewed literature: scientific journals, books and conference proceedings
 - Created by extracting records from child databases and enhanced with their cited references
 - Bibliometric data poor maintained for prior 1996 records

Principles Underlying Journal Impact Factor

- Journal Impact Factor (JIF): Clarivate Analytics (which used to be a part of Thomson Reuters till 2016) publishes JIF in a periodic journal called Journal Citation Reports (JCR) to be used as a quality measure.
- JIF is a part of Web of Science (WoS) database. Clarivate owns and operates a collection of subscription-based services focused largely on analytics which includes Web of Science (WoS).
- JIF is calculated by as the average of the sum of the citations received in a given year to a journal's previous two years of publications (linked to the journal, but not necessarily to specific publications) divided by the sum of "citable" publications in the previous two years.

Computation of Journal Impact Factor

- Consider a journal, SRMIST journal that has been published in 2022, 2023 and 2024.
- Total number of articles published in 2022 = 130
- Total number of articles published in 2023 = 170SRMTotal number of articles published in 2022 and 2023, say $A = 300$
- So, SRMIST journal had a total of 300 publications in 2022 and 2023 which had a potential to be cited as a reference by articles that appeared subsequent years, say in 2024 in various journals.
- Please note that there is a possibility that some of the 300 publications are poor quality that would not attract any citations in 2024. Also, some of the articles might be extraordinary enough to attract very high citations in the future years. Hence, different articles will attract different number of citations. All these citations put together regardless of which article attracted how many citations go onto define the magnitude of B.

- Let the total number of citations garnered from various journal (SRMIST as well as other journals) in 2024 by the 300 articles that appeared in SRMIST Journal in 2022 and 2023, say $B = 600$
- Please note the citations in 2024 is not restricted to SRMIST journal and maybe from other journals as well.
- Then JIF for SRMIST journal in 2024 is $B/A = 600/300 = 2$

$$IF \text{ of Year} = \frac{\text{\# of citations to Journal X in the past two years}}{\text{\# of total citable items by Journal X in the past two years}}$$

- Please note that IF for 2024 can be calculated only in 2025 after all journal issues have been published and their citations have been populated in the database.
- The citations garnered in the last two years is a measure of the significance of the journals and is reflected in its impact factor. So, editors try to maximize B value in the expression!

International Journal of Hydrogen Energy

COUNTRY United Kingdom Universities and research institutions in United Kingdom Media Ranking in United Kingdom	SUBJECT AREA AND CATEGORY Energy - Energy Engineering and Power Technology - Fuel Technology - Renewable Energy, Sustainability and the Environment Physics and Astronomy - Condensed Matter Physics	PUBLISHER Elsevier Ltd	SJR 2024 1.685 Q1 H-INDEX 285
PUBLICATION TYPE Journals	ISSN 03603199	COVERAGE 1976-2025	INFORMATION Homepage How to publish in this journal Activate Windows Go to Settings to activate Windows.

Merits of Journal Impact Factor

- Authors prefer to publish in higher impact factor journals because:
 - Higher citations rate means article has higher chances of getting cited
 - Higher reading rate means article gets read by more researchers
 - More recognition in terms of scholarly contribution
- JIF provides an input about how frequently an average article in a journal has been cited in a particular year
- JIF informs us about the prestige of a journal as a whole
- Editors use it to judge the popularity of their journals and strive to enhance the JIF by choosing good articles that have potential to garner more citations and eliminate ones that might not.
- In libraries, it is a tool used for management of library journal collections (unsubscribe from low JIF ones)
- In market research, publishers use it as a quantitative evidence about the popularity of their journals

Demerits of Journal Impact Factor

- JIF is an average statistic and not all articles are equally cited. So, an article having fewer citations published in a high IF journal is given more importance than a highly cited article in a low IF journal.
- Only includes "citable" articles in the denominator of the equation, i.e., articles and reviews
- Only a limited subset of journals is indexed by Clarivate Analytics
- While calculating JIF, self citations are also counted.
- Tell us nothing concrete about any specific paper or specific author
- Review articles are generally highly cited. Publication of more number of review papers in a journal can greatly affect the journal's IF.
- IF depends on the size of the field concerned. A larger field size would draw more citations than the one having a small size.
- Biased toward English language journals and JCR is also expensive

Journal Immediacy Index

- Journal Immediacy Index (JII) is also reported in JCR and is a measure about how topical or state-of-the-art the topic of a published article might be.
- This is measured by how quickly a published article in a given journal is cited in the same calendar year.
- It is the ratio of total citations for a journal's articles published during a calendar year divided by the total number of articles published in the same calendar year.

Journal Immediacy Index ⓘ

Cites in 2007 to articles published in 2007 = 2343

Number of articles published in 2007 = 366

Calculation: $\frac{\text{Cites to current articles}}{\text{Number of current articles}} = \frac{2343}{366} = 6.402$

- Journals that publish only review articles such as "Chemical Reviews" are likely to garner more citations in the same year due to wide applicability of the review articles.

3. Author-Level Metrics

These tools evaluate an individual researcher's productivity and impact in their field.

Examples:

h-index: An author has an h-index of n if n of their papers have at least n citations each. Balances productivity and citation impact.

i10-index (Google Scholar): Number of publications with at least 10 citations.

Citation count: Total number of times a researcher's publications have been cited.

ResearchGate Score / RG Score: Combines publications, citations, Q&A, and followers to estimate scientific reputation on ResearchGate.

ORCID and Scopus Author ID: Help track and distinguish author contributions across platforms.



John Hopfield

FOLLOW

Professor, [Princeton University](#)

Verified email at princeton.edu - [Homepage](#)

[Neural Networks](#) [AI](#) [Neuroscience](#) [Systems Biology](#) [Semiconductor Physics](#)

Journals' Rankings

BETA ?

[Report a Bug](#) · [Options](#) · [Vote for Next Features](#)

Q1
50% 2

NA ?
50% 2

— based on 4 most cited papers in the last 10 years ?

ExCITATION.tech

TITLE

CITED BY

YEAR

[Neural networks and physical systems with emergent collective computational abilities.](#)

29001

1982

JJ Hopfield

3.414 Proceedings of the national academy of sciences 79 (8), 2554-2558

[Neurons with graded response have collective computational properties like those of two-state neurons.](#)

9914

1984

JJ Hopfield

3.414 Proceedings of the national academy of sciences 81 (10), 3088-3092

["Neural" computation of decisions in optimization problems](#)

9084

1985

JJ Hopfield, DW Tank

Cited by

[VIEW ALL](#)

All

Since 2020

Citations

93885

15356

h-index

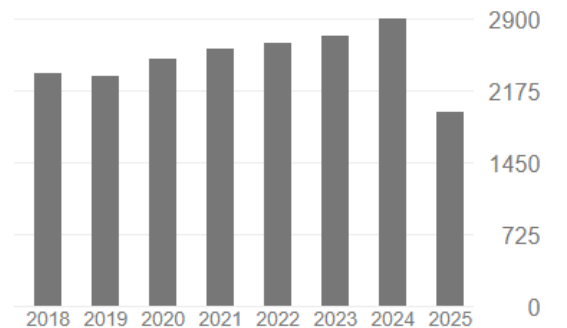
95

33

i10-index

177

76



Public access

[VIEW ALL](#)

0 articles

2 articles

not available

available

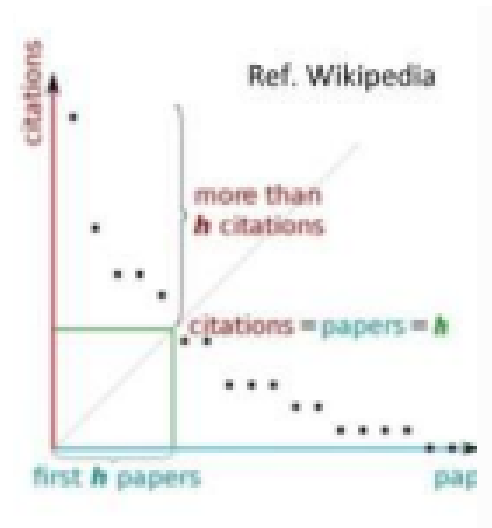
Based on funding mandates

h-index (Hirsch index):

- The h-index was suggested in 2005 by Jorge E. Hirsch, a physicist at UC San Diego, USA and hence also called as "Hirsch Index".
- It is a metric for evaluating the cumulative impact of an author's scholarly output and performance
- It combines an assessment of both quantity (number of papers) and quality (impact, or citations to these papers)
- It is automatically calculated by citation indexing agencies such as Web of Science, Scopus, Google Scholar etc. But, it can also be calculated manually.
- The index can also be applied to the productivity and impact of a scholarly journal as well as a group of scientists, such as a department or university or country

- The h-index is defined as the maximum value of h such that the given author/journal has published " h " papers that have each been cited at least " h " times.
- For example consider the tabulation for "rank of article" vs. "decreasing number of citations for a particular author."
- The articles are listed in decreasing order of number of citations and then plotted for serial number of article vs. number of citations as shown in the plot.
- A diagonal line is drawn across the plot that shows that the h-index is 5. This can also be inferred from the table, where the article serial number and number of citations match

Articles	Citations
1	46
2	25
3	12
4	11
5	5 = H - Index
6	4
7	2
8	1



Factors influencing h-index

- The h-index of a particular author may vary from one database to another, depending on the coverage of each database.
- Co-publishing can elevate the h-index, because it does not take the order of authorship into account.
- Time helps to increase the h-index; young researchers generally have a low h-index.
- The subject area can have an effect on the h-Index: a researcher who publishes on a trendy subject will have a higher h-index than another one who publishes on a niche topic.
- The language affects h-index; with English receiving more citations.
- Open access publishing increases the visibility of articles and increase their chance to be cited.
- Promotion of publications (blog posts, conferences etc.) helps to make articles more visible and increases their chance of being cited.
- h-index does not eliminate self-citations!

Source Normalized Impact per Paper (SNIP)

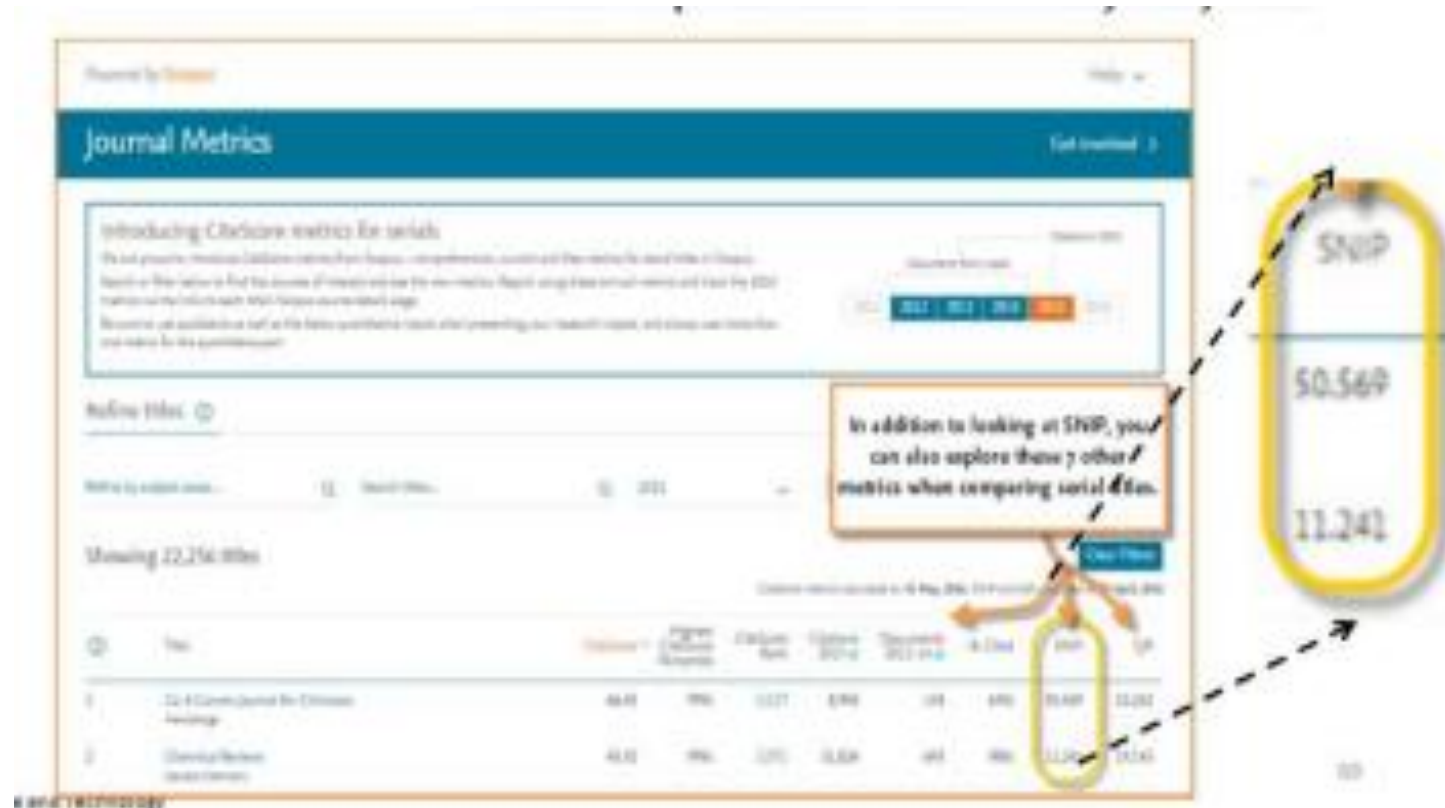
- SNIP is a factor released in 2012 by Elsevier based on Scopus to estimate impact of a journal article.
- The measure is calculated as $SNIP = RIP / (R/M)$, where RIP = Raw Impact per Paper, R = Citation Potential and M = Median database citation potential.
- SNIP corrects for differences in citation practices between scientific fields, thereby allowing for more accurate between-field comparisons of citation impact.
- The main differences between the SNIP and JIF are:

Differences between SNIP and JIF		
	SNIP	JIF
Database based on	Scopus	Web of Science
Field Differences taken into account	Corrected	No correction
No. of Year of Cited Publications	Three	Two
Citation Sources	Selected sources only	All sources
Citations Referred to	Selected documents only	All types of documents

Calculation of SNIP

- SNIP can be obtained from www.journalindicators.com. It can also be obtained from the Scopus' Journal Metric webpage as shown below. It can also be calculated as shown below.

$$SNIP \text{ of journal} = \frac{\text{Journal's citation count per article}}{\text{Citation potential in the subject field}}$$



g-Index - Introduction and Computation

- The g-index also known as Egghe's g-index was introduced in 2006 as an improvement of the h-index of Hirsch to measure the global citation performance of a set of articles.
- If this set is ranked in decreasing order of the number of citations that they received, the g-index is the (unique) largest number such that the top g articles received (together) at least g^2 citations.
- Consider a list of articles with **descending number of citations** as shown in the next slide. For the sake of brevity, only certain parts that show how the g-index is calculated from the list are presented. The "Article No." is the number for listing the publications considered and starts from number 1. The "Article No. Squared" is the square of the "Article No.". The "Citation, Citn" is the number of citations received for each publication respectively.

Computation of g-Index

- The " $Cit_n + \sum Cit_{n-1}$ " is the cumulative sum of the citations starting from number 1 to n. At the start of the table, the " $Cit_n + \sum Cit_{n-1}$ " value > "Article No. Squared" and this reverses by the end of the table.
- The g-index corresponds to the article number in the row where this reversal occurs, ie. "Article No Squared" number exceeds the " $Cit_n + \sum Cit_{n-1}$ " number. In the example cited, g-index is 54

Article No.	Article No. Squared	Citation (Cit_n)	$Cit_n + \sum Cit_{n-1}$
48	2304	-	-
49	2401	-	-
50	2500		2972
51	2601	3	2975
52	2704	3	2978
53	2809	2	2980
54	2916	2	2982
55	3025	1	2983
56	3136	1	2984
57	3249	1	2985
58	3364	0	2985
59	3481	-	-
60	3600	-	-

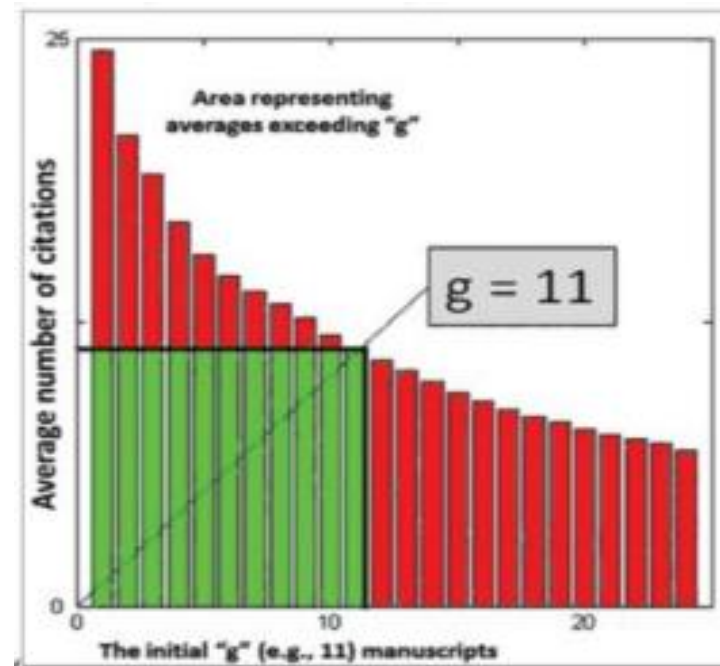
g index – highest number of papers that together received g^2 or more citations

Publication where * $Cit_n + \sum Cit_{n-1}$ * number > *Article No. Squared * number

g-Index - Significance

Some factors to be considered for g-index.

- The g-index gives more weight to highly-cited articles (whereas h-index is insensitive to it).
- The g-index will always be equal to or greater than the h-index.
- The g-index can be checked through "Publish or Perish" - a free downloadable software for academic citations analysis, URL: <https://harzing.com/resources/publish-or-perish>.



i10-Index - Introduction and Computation

- The i10-index was created by Google Scholar in 2011.
- The i10-Index for a particular author is the number of publications with at least 10 citations.
- It is accessible only from Google Scholar Citations profile of the author.
- i10-index is a very simple measure proposed by Google Scholar, and is another way to help gauge the productivity of a scholar.
- Therefore, if someone has an i10-index of 32, it means that the person has 32 papers with citations more than 10. In reality, many papers may have much more than 10 citations (say 100 or even more), but reference frame is at least 10

- Advantages of i10-IndexSerial

- Very simple and straight forward to calculate
- My Citations in Google65Scholar is free and easy to use

- Disadvantages of i10-Index

- Used only in Google Scholar
- Minimum requirement is 10 citations per article, so there is no weightage for articles with more than 10 citations.

Serial No.	Number of Citation
1	75
2	68
3	65
4	60
5	45
6	15
7	12
8	9
9	5
10	0

Meta-Analysis of Research Findings

Meta-analysis is a quantitative technique used to combine and analyze results from multiple independent studies on the same topic to arrive at a comprehensive conclusion.

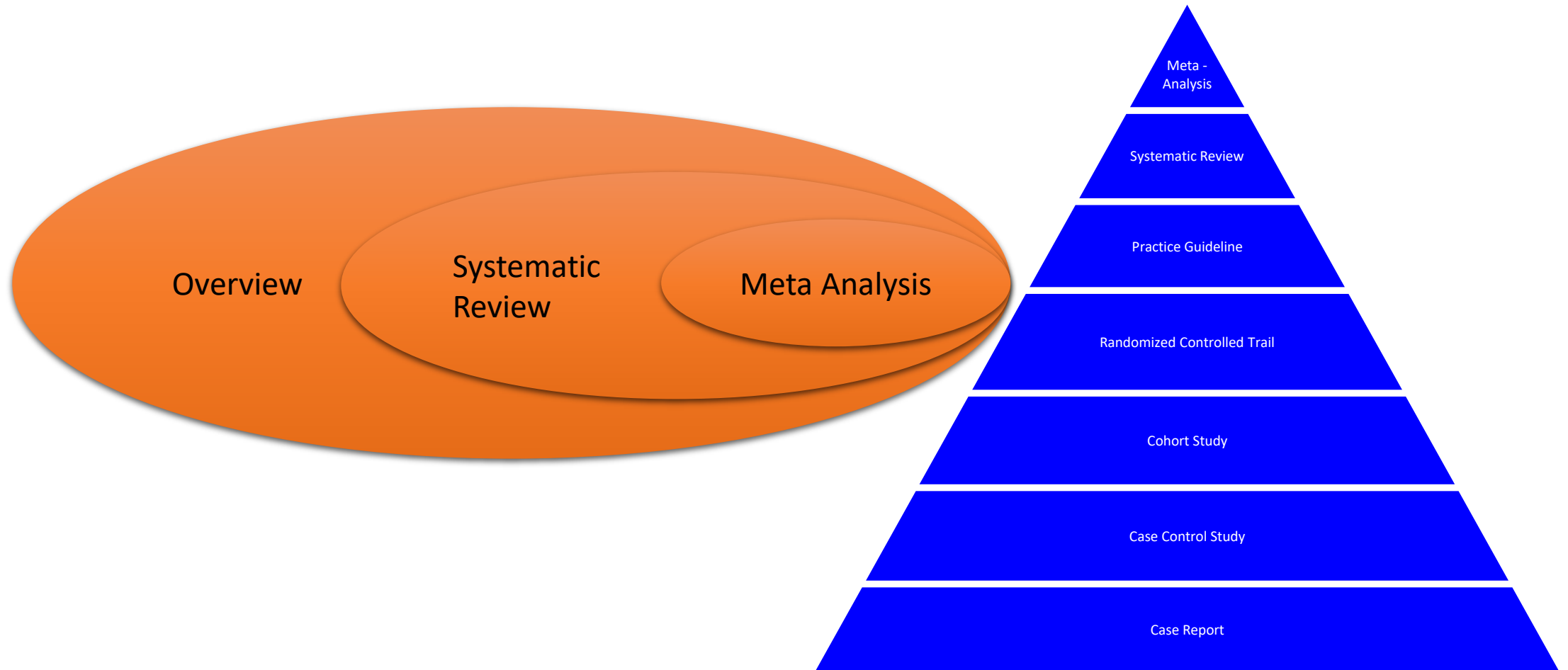
Key Features:

- Systematic approach: Gathers studies using predefined criteria.
- Statistical aggregation: Combines effect sizes or summary statistics.
- Increases power: Enhances reliability and generalizability of results.
- Identifies trends: Detects patterns, inconsistencies, or gaps across studies.

Steps Involved:

1. Define research question and inclusion criteria.
2. Collect and screen relevant studies.
3. Extract data (e.g., effect sizes, sample sizes).
4. Perform statistical analysis (e.g., weighted averages, forest plots).
5. Interpret and report findings.

Meta-analysis is often part of the systematic review process. It is used when you are comparing multiple studies that looked at similar relationships.



Literature

Definition and General Characteristics

- A body of work that has been distributed, written or published on a given topic or concept
- Can be in multiple formats, depending on the academic subject or discipline
- Various intents: inform, support, persuade, update, or dispute

The literature consists of the following :

- Major works : key or seminal publications on a topic
- Other works or publications that build upon or respond to the major works.
- The literature is a continuously evolving network of scholarly works that interact with each other.

Types and Sources of Literature

- Literature falls into three categories : Primary, Secondary and Tertiary Sources

PRIMARY LITERATURE



- Original research and / or new scientific discoveries
- Immediate results of research activities
- Often includes analysis of data collected in the field or laboratory

EXAMPLES

- Original research published as articles in peer – reviewed journals
- Dissertations
- Technical reports
- Conference proceedings

SECONDARY LITERATURE



- Summarizes and synthesizes primary literature
- Usually broader and less current than primary literature

EXAMPLES

- Literature review articles
- Books

Since most information sources in the secondary literature contain extensive bibliographies, they can be useful for finding more information on a topic

TERTIARY LITERATURE



- Summaries or condensed versions of materials
- Usually with references to primary or secondary sources
- Good place to look up facts or get a general overview of a subject

EXAMPLES

- Textbooks
- Dictionaries
- Encyclopedias
- Handbook

Primary Sources of Literature

- 1. Primary Sources: first-hand accounts of events, practices, or conditions being researched.



- It also includes many original, creative works or artifacts.



Caliban's Rebellion (2008)

Image credit: Shehal Joseph at flickr.com. Permission by CC by 2.0

Primary Source Examples:

Humanities :

- Creative works (original literature, fiction or non-fiction, poetry, film, performances)
- Music scores handwritten by the composer(s)
- Diaries
- Regalia or original artefacts
- Magazine or newspaper articles
- Autobiographies
- Memoirs
- Oral histories
- Field notes
- Raw data
- Statues and laws

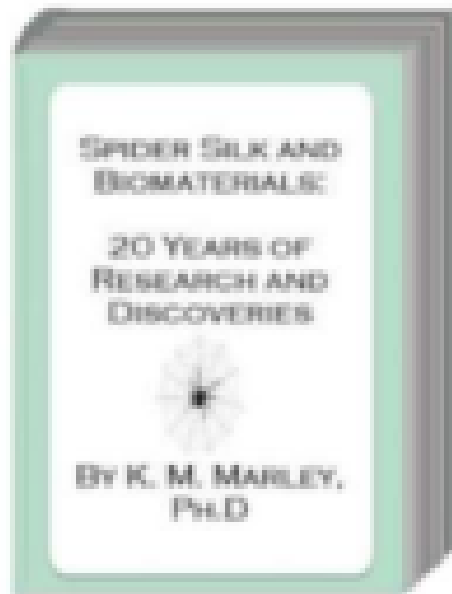
Sciences/Health Sciences :

- Original research (articles, reports, pre-prints).
- Conference papers
- Interviews
- Dissertations & theses
- Patents & inventions
- Technical reports
- Lab notebooks
- Raw data

Social sciences tend to use a combination from these broad disciplines.

Secondary Sources of Literature

- Secondary Sources: sources that discuss, study, or comment on information from primary sources



2020-21, Open Access Research Journal for
Humanities, Arts,
15(1), 1-10 (2021)

<https://www.researchjournalforhumanities.com>
DOI: 10.21969/rjh.15101

THE PERFORMANCE STUDIES PARADIGM

Aurora Sengul

Abstract: *The aim of this paper is to show how performance can be considered as paradigmatic practice significant for understanding of the contemporary art not only in the field of theatre and performance art, but also all other artistic and cultural practices as well as a number of non-artistic practices. I will develop my argument based in on the Richard Schechner's understanding of performance as a collection of interpretative tools for studying broad spectrum of social phenomena, where the objects of research don't have to be clarified according to the medium, are bound to specific disciplines and Jon Mckenzie's understanding of performance as auto-historic formation of knowledge and power where performance is established as a general paradigm of contemporary society. In the central discussion I will refer to the significance of the concept of liminality.*

Keywords: *performance, liminality, knowledge, power, contemporary art*

Secondary Source Examples:

Humanities:

- Biographies
- Literary criticism
- Monographs written about a topic
- Reviews on creative works
- Music scores that have been published or changed
- Bibliographies
- Law reviews
- Journal articles (criticism and interpretation)

Sciences/ Health Sciences :

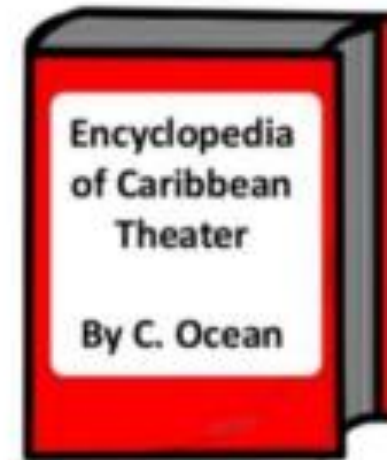
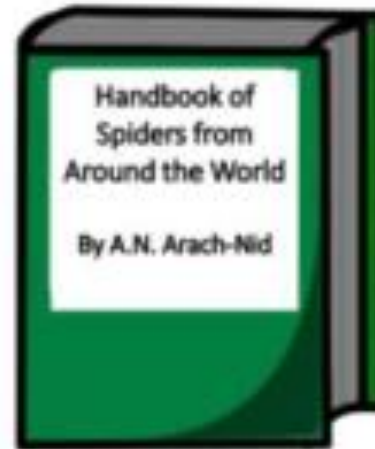
- Literature reviews
- Data analysis or refined compilations
- Reviews (systematic or meta-analysis) Opinion pieces*
- Letters to the editor
- Articles in trade journals (non-research)
- Text books and books

Tertiary Sources of Literature

- Tertiary Sources: sources that utilize and distill information from primary and secondary sources.

Reference works such as :

- Handbooks
- Standards
- Almanacs
- Encyclopedias
- Manuals
- Pathfinders (Lib Guides)



1. What is a Literature Review?

A literature review is a critical summary of existing research on a particular topic. It involves identifying, reading, summarizing, grouping, comparing, and evaluating previous studies and scholarly articles relevant to the research question. The goal is to understand the current state of knowledge, identify gaps, and position the new research within the existing body of work.

2. Grouping of Literature

Grouping is the process of organizing reviewed literature based on themes, theories, methodologies, or chronology. For example, if studying hydrogen storage in Mg alloys, papers could be grouped under:

- Alloy composition studies
- Processing methods
- Machine learning-based prediction
- Kinetics and thermodynamics

This thematic arrangement helps in comparing similar works and observing trends or contradictions.

3. Analysis of Literature

Analysis involves a critical examination of the literature, not just summarizing it. You assess:

- The objectives and findings
- Methodologies used
- Strengths and limitations
- Relevance to your own research

By analyzing, you recognize which works are landmark studies, which findings are widely accepted, and where disagreements or gaps exist.

Research Methods

In research methodology, research methods refer to the techniques or procedures used to collect and analyze data related to a research problem. Identifying the appropriate method is crucial as it ensures that the research question is addressed accurately and scientifically. The choice depends on the nature of the problem, objectives, available data, and resources.

2. Classification of Research Methods

Broadly, research methods are classified into qualitative, quantitative, and mixed methods:

- Qualitative methods focus on exploring ideas, understanding meanings, and gaining insights (e.g., interviews, focus groups).
- Quantitative methods emphasize numerical data, measurement, and statistical analysis (e.g., surveys, experiments).
- Mixed methods combine both, giving a more holistic understanding of the problem.

The selection depends on whether the study seeks to explain, explore, or predict phenomena.

3. Experimental Method

This method involves manipulating one or more variables to observe their effect on another variable. It is commonly used in natural sciences and engineering, such as testing the effect of temperature on hydrogen absorption in metal hydrides. The key strength is control over variables, allowing for cause-effect relationships. It requires a lab setup, controlled conditions, and often random sampling.

4. Descriptive Method

Descriptive research aims to systematically describe a situation, event, or population. It doesn't involve manipulation of variables. Tools include surveys, observations, and case studies. For example, surveying researchers to understand trends in machine learning usage in materials science. It's useful for developing hypotheses and getting a broad view of a phenomenon.

5. Analytical Method

This method involves using already available information and analyzing it to make a critical evaluation. For example, re-analyzing published data on Mg-based alloys using a new algorithm to identify patterns. Techniques include meta-analysis, data mining, and statistical re-evaluation. It is often used in secondary research.

6. Historical Method

This method involves investigating past records and events to understand trends, origins, or development of certain phenomena. It is widely used in social sciences and policy research. For instance, tracing the development of hydrogen storage technologies over decades to identify gaps and future directions.

7. Case Study Method

The case study method focuses on an in-depth analysis of a single case or a small group of related cases. This could involve studying a specific hydrogen storage alloy and all its fabrication, performance, and optimization data in detail. It is valuable when the research question requires a detailed contextual understanding.

8. Survey Method

The survey method is used to collect data from a large population using questionnaires or structured interviews. It is best for gathering opinions, attitudes, or factual information at scale. In materials research, this could include expert opinion surveys on future alloying strategies for hydrogen storage.

9. Simulation and Modelling Method

Increasingly used in science and engineering, simulation and modeling involve creating mathematical or computational models to mimic real-world systems. For example, machine learning models, finite element models, or thermodynamic simulations to predict hydrogen diffusion or storage capacity.

10. Field and Laboratory Methods

These include direct measurements, sampling, and experiments carried out either in the lab or in real-world settings. Lab methods provide precision; field methods offer realistic conditions. A study may use both—testing a material's hydrogen uptake in the lab and validating it under industrial conditions.

Intellectual Property Rights (IPR)

Definition and Motivation for IPR

- Intellectual property rights are legal rights that provide creators protection for original works, inventions, or the appearance of products, artistic works, scientific developments etc.
- This incentivizes discovery and creativity by providing creators with an opportunity to profit from the value of their innovative work
- The IP rights are important because they can:
 - Used to differentiate a business from its competitors
 - Be sold or licensed providing an important revenue stream
 - Offer customers something innovative that is not available with other businesses etc.
 - This encourages innovation, competitiveness, knowledge economy development, jobs and economic growth and prosperity

Broad Classification of IPR

- Intellectual property is broadly categorized into two:
 - i. Industrial Property-Patents, trademarks and designs-eg. drug molecule patent, coca cola trademark and bottle design

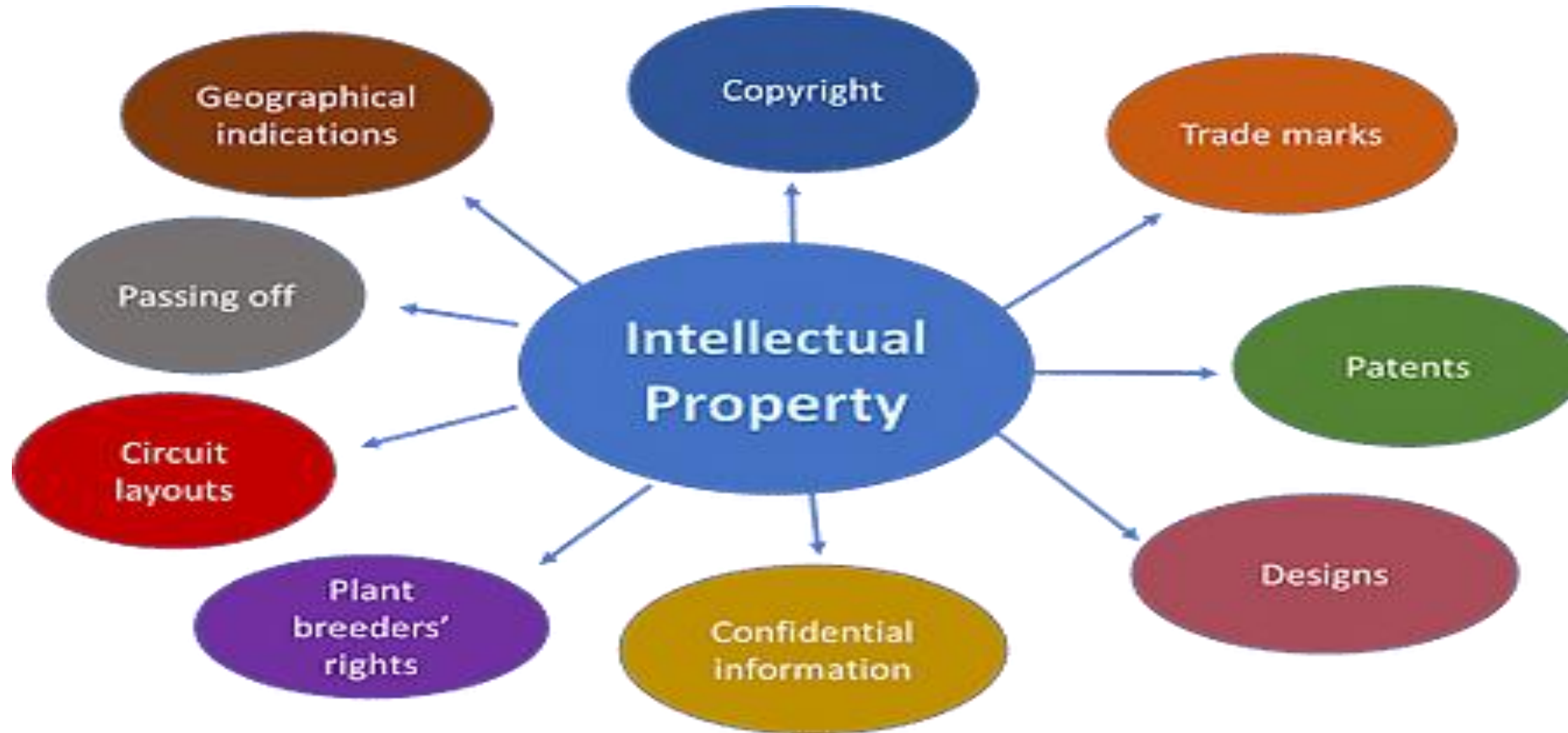


- ii. Literary and Artistic Property-copyright and related rights- eg. Shakespeare's literary works, Computer codes, Video recordings, Musical works, Website design, Pictures etc



Intellectual Property Rights (IPR)

Intellectual Property Rights (IPR) are legal protections granted to creators and inventors for their unique creations and innovations. These rights allow individuals or organizations to control the use of their intellectual creations—be it artistic works, inventions, brand symbols, or industrial designs. The primary aim of IPR is to encourage creativity and innovation by ensuring that creators can benefit financially from their work while preventing unauthorized usage.

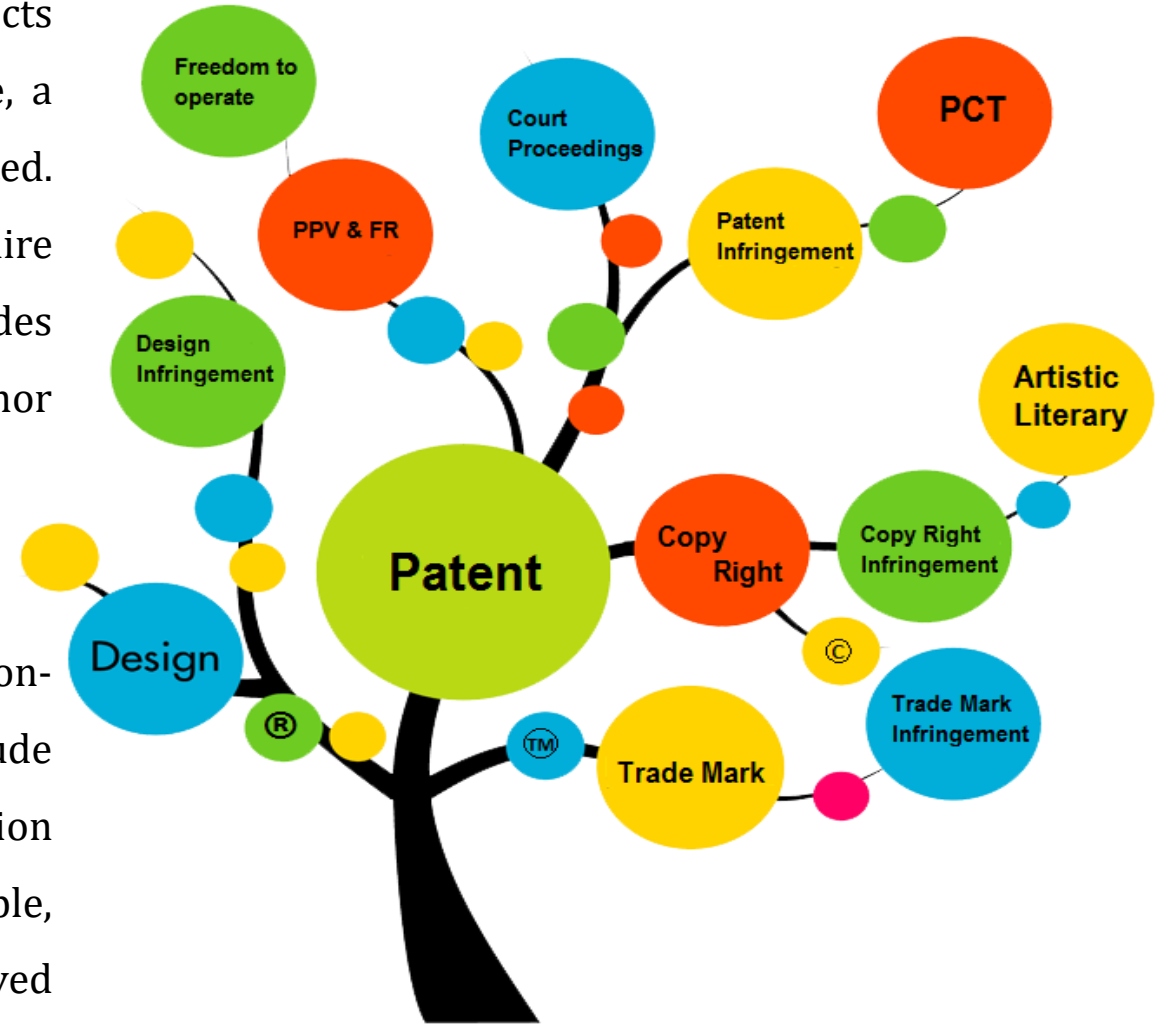


Copyright

Copyright refers to the exclusive legal right given to the creator of original literary, artistic, musical, and dramatic works. It protects expressions of ideas rather than the ideas themselves. For example, a research paper, a painting, a novel, or software code can be copyrighted. Copyright arises automatically upon creation and does not require registration in many jurisdictions, although registration provides stronger legal evidence. It typically lasts for the lifetime of the author plus 60 or 70 years.

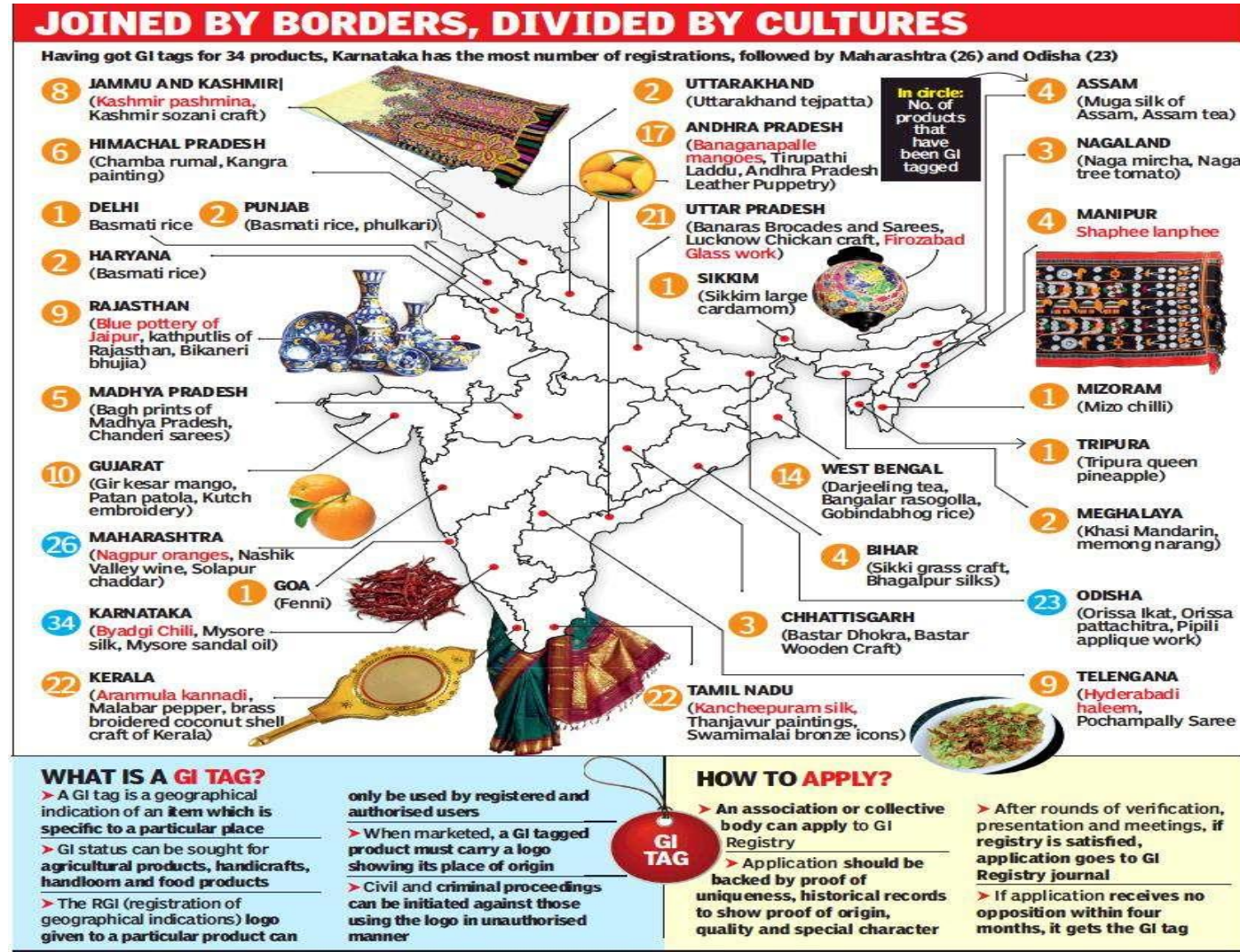
Patent

A patent is a right granted to an inventor for a novel, useful, and non-obvious invention or process. Patents allow the inventor to exclude others from making, using, or selling the invention without permission for a certain period—usually 20 years from the filing date. For example, in hydrogen storage research, a new Mg-based alloy with improved storage capacity could be patented. Patents must be applied for and granted by a government body, and they require complete technical disclosure of the invention.



Geographical Indication (GI)

A Geographical Indication (GI) is a type of IPR that protects products which have a specific geographical origin and possess qualities or a reputation due to that origin. Examples include Darjeeling tea, Basmati rice, and Kanchipuram silk. GIs help preserve traditional knowledge and practices and promote rural economic development. Only authorized users residing in the geographic territory can use the name protected under GI.



Industrial Designs

Industrial designs refer to the aesthetic aspects of a product, such as its shape, configuration, pattern, or ornamentation. They protect the visual design of objects that are not purely utilitarian. For instance, the unique design of a smartphone, furniture, or packaging can be protected through an industrial design right. These rights typically last between 10 to 15 years, depending on the country and can be renewed in some cases.

Technology Transfer

Technology transfer is the process of sharing or disseminating technology from one organization or country to another. It often happens through licensing agreements, research collaborations, or commercialization of patents. Universities and R&D institutions often engage in technology transfer by licensing innovations to industries for product development. It plays a vital role in translating research outcomes into real-world applications, especially in sectors like pharmaceuticals, materials science, and renewable energy.

IPR Type	Portal	Key Steps	Time
Copyright	copyright.gov.in	Apply → Upload work → Pay → Objection check → Register	2–3 months
Patent	ipindia.gov.in	Search → File (Form 1,2) → Exam (Form 18) → Response → Grant	2–4 years
Trademark	ipindia.gov.in	Search → File TM-A → Pay → Journal → Register	6–12 months
GI Tag	ipindia.gov.in	File GI Form → Submit proof → Review → Publish → Register	6–12 months
Design	ipindia.gov.in	Submit drawings + Form 1 → Pay → Examination → Register	6–12 months

ipindia.gov.in



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Ministry of Commerce & Industry,
Government of India



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Industry and Internal Trade
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Practice in Research Methodology

Focus Areas:

- 1. Identifying Research Topic
- 2. Formulating Research Statement
- 3. Conducting Literature Review

- Objective: Apply theoretical knowledge to hands-on research planning

Steps to Identify a Research Topic

Choose Broad Area of Interest

Preliminary Literature Review

Identify Gaps / Problems

Brainstorm Possible Topics

Evaluate Feasibility

Formulate Research Problem & Objectives

Formulating Research Statement

Definition:

A concise description of what the research will study

Characteristics of a Good Statement:

- - Specific & Clear
 - - Researchable & Feasible
 - - Addresses a Gap or Problem
 - - Aligned with Objectives
-
- Example: Investigating the effect of nano-coatings on the corrosion resistance
 - of magnesium alloys for biomedical implants.

Conducting a Literature Review

Steps:

- 1. Keyword Search (Google Scholar, Scopus, Web of Science)
- 2. Backward Search – References of selected papers
- 3. Forward Search – Who cited those papers later
- 4. Group & Compare studies by theme, method, or results
- 5. Analyze for gaps, contradictions, and opportunities

Literature Review – Presentation

Organize findings into:

- Tables & Charts (Summary of methods & results)
- Concept Maps (Connecting studies)
- Conclude: Identify trends, gaps, and research needs
- Outcome: Clear research direction for your project